DATA COMMUNICATION
MODULE -1
INTRODUCTION

DATA COMMUNICATION

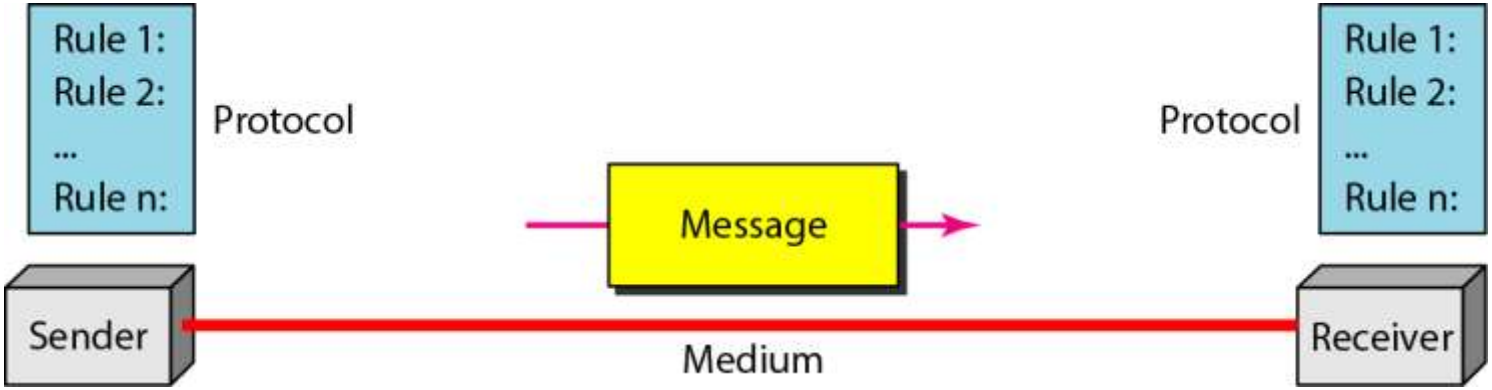
Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Fundamental characteristics[Effectiveness of DC]:

1. *Delivery-correct destination*
2. *Accuracy-deliver data accurately*
3. *Timeliness- delivery data in timely manner*
4. *Jitter-variation in packet arrival time.*

Components of a data communication system

- **Message-** information to be communicated
- **Sender-** send the data message
- **Receiver –** receives the message
- **Transmission Medium-** physical path by which message travels from sender to receiver.
- **Protocol-** set of rules govern the data communications



Data Representation

- Text
  - Represented as bit pattern (sequence of bits 0s or 1s)
  - Different set of bit pattern used to represent symbols or characters. Each set is called code Process of representing symbols is called encoding
  - Ex: ASCII,UNICODE

- Numbers
  - Represented as bit pattern
  - Directly converted to binary form

- Audio
  - Recording or broadcasting of sound or music.
  - Continuous not discrete

Data Representation (Cont.…)

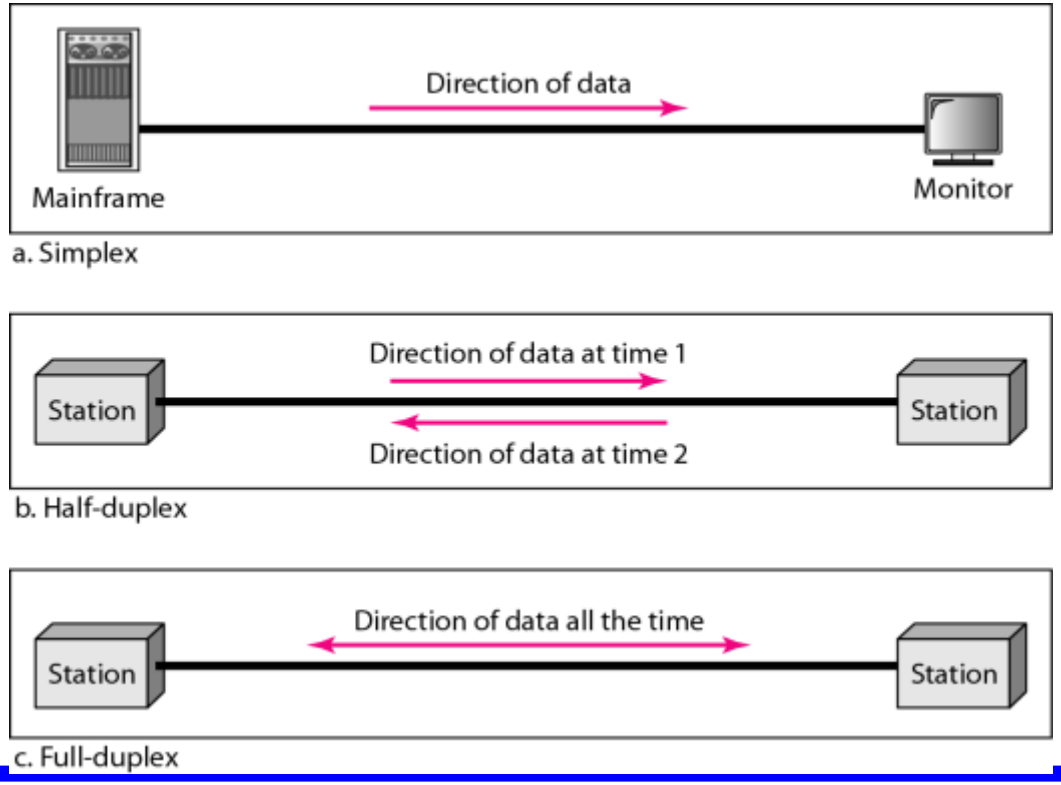
- Video
  - Recording or broadcasting of picture or a movie
  - Produced as :
    - Continuous entity [TV camera]
    - Combination of images-discrete entity

- Images
  - Represented as bit pattern
  - Image is divided into matrix of pixels(smallest element of an image)
  - Each pixel is assigned a bit pattern (size and value of pattern depend on image)
  - Ex: black and white dots (chessboard) -1 bit pattern is enough to represent a pixel, gray scale- 2 bit pattern.
  - Several methods to represent colour images : RGB,YCM

Data flow (simplex, half-duplex, and full-duplex)

Three ways of data-flow between 2 devices :

1) Simplex
2) Half-duplex
3) Full-duplex



NETWORKS

A network is a set of devices (often referred to as nodes) connected by communication links.

A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

A link can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

NETWORK

- A network is defined as a set of devices interconnected by communication-links. This interconnection among computers facilitates information sharing among them.

- Computers may connect to each other by either wired or wireless media. Often, devices are referred to as nodes.

- A node can be any device capable of sending/receiving data in the network.
  For example: Computer & Printer

- The best-known computer network is the Internet

Network Criteria

- **Performance**
  - **Measured using:**
    - **Transit time**: time taken to travel a message from one device to another.
    - **Response time**: time elapsed between enquiry and response.
  - **Depends on following factors:**
    - Number of users
    - Type of transmission medium
    - Efficiency of software
  - **Evaluated by 2 networking metrics:**
    - Throughput (high)
    - Delay (small)

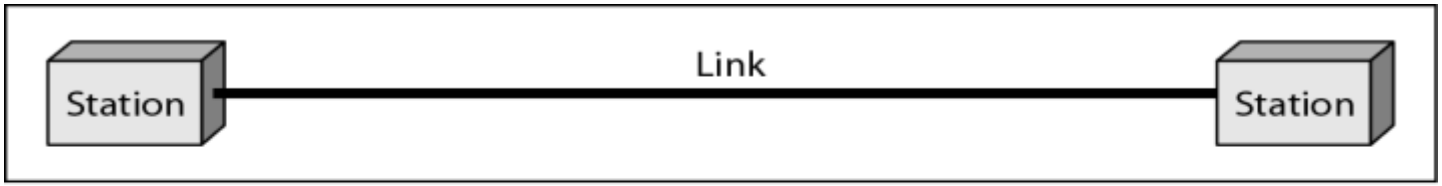
Network Criteria (cont..)

- **Reliability**
  - **Measured by**
    - Frequency of failure.
    - Time taken to recover from a network failure.
    - Network robustness in a disaster.

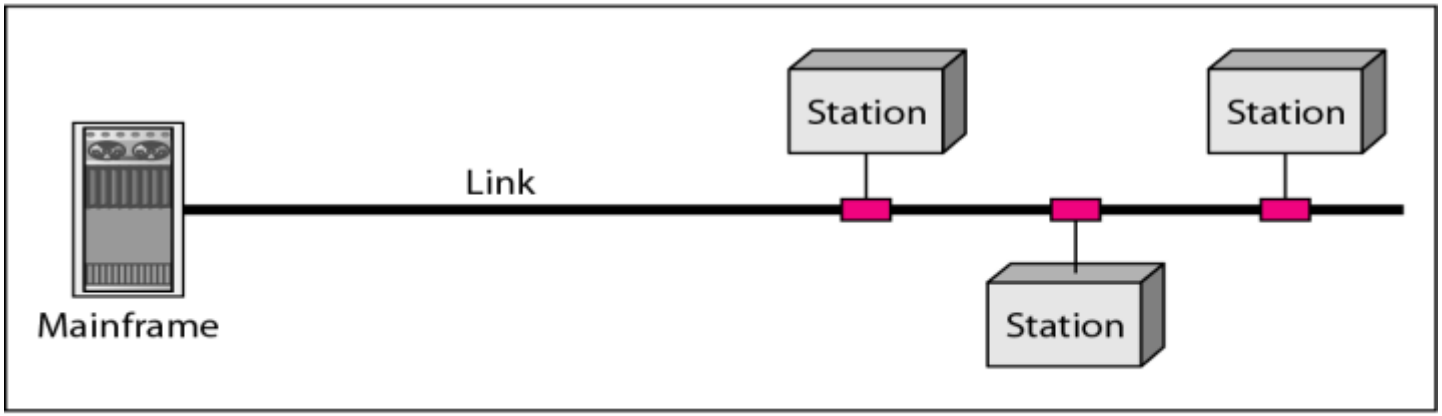
- **Security**
  - **Protecting** data from unauthorized access, damage and development.
  - **Implementing** policies and procedures for recovery from breaches and data losses.

Physical Structures

- **Type of Connection**
  - **Point to Point - single transmitter and receiver**
  - **Multipoint - multiple recipients of single transmission**



a. Point-to-point



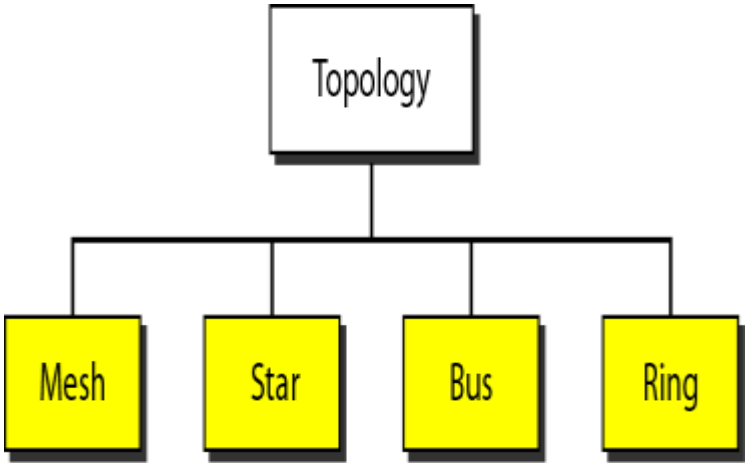
b. Multipoint

Physical Structures (Cont..)

- **Physical Topology:**

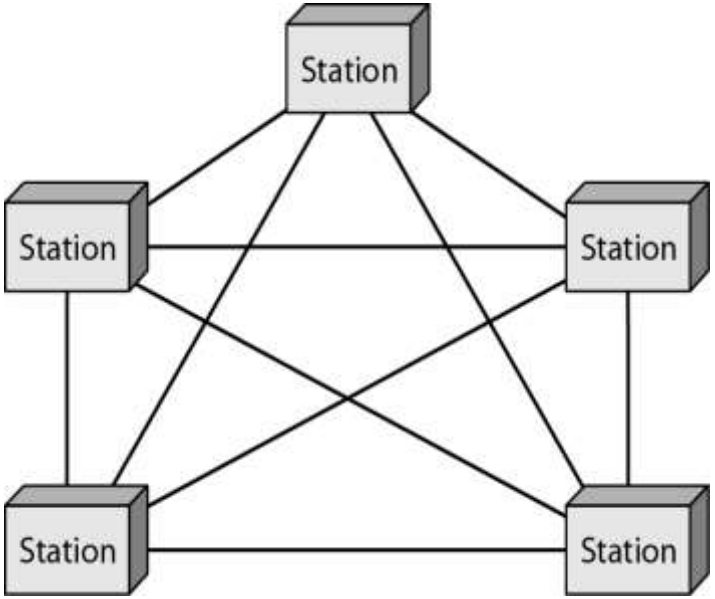
  - **Way in which network is laid out physically.**
  - **Two or more devices form link and two or more links form a topology.**

  - ***Basic topologies:***
    - *Mesh*
    - *Star*
    - *Bus and*
    - *Ring*



Mesh Topology

- All devices are connected to each other
- Dedicated point-point link between all devices
- “n(n-1)” physical channels to link “n” device.
- For „n‟ nodes
  - n(n-1) physical links
  - n(n-1)/2 duplex mode links
- Every device have (n-1) I/O ports to be connected to other (n-1) devices.



Fully connected Mesh Topology

Mesh Topology (cont..)

Advantages:

Advantages:

1) Congestion reduced: Each connection can carry its own data load.
2) Robustness: If one link fails, it does not affect the entire system.
3) Security: When a data travels on a dedicated-line, only intended-receiver can see the data.
4) Easy fault identification & fault isolation: Traffic can be re-routed to avoid problematic links.

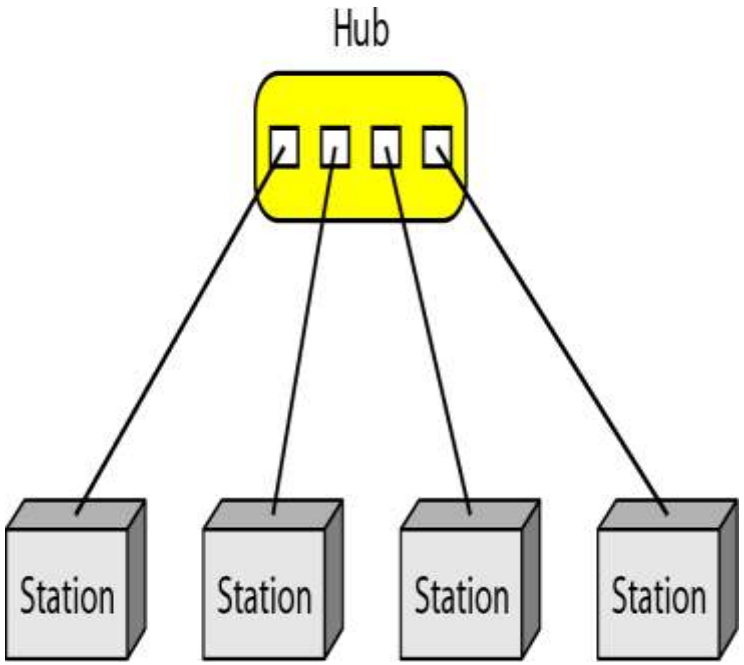
Disadvantages:

1) Difficult installation and reconfiguration.
2) Bulk of wiring occupies more space than available space.
3) Very expensive: hardware required to connect each link is expensive.

practical example: connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

Star Topology

- Point to Point connection
- All the devices are connected to a central controller called a hub
- Dedicated point-to-point link between a device & a hub.
- The devices are not directly linked to one another. Thus, there is no direct traffic between devices.
- The hub acts as a junction:
  - If device-1 wants to send data to device-2,
  - the device-1 sends the data to the hub,then the hub relays the data to the device-2.



A star topology connecting four stations

Star Topology (cont..)

Advantages:

1) Less expensive: Each device needs only one link & one I/O port to connect it to any devices.

2) Easy installation & reconfiguration: Nodes can be added/removed w/o affecting the network.

3) Robustness: If one link fails, it does not affect the entire system.

4) Easy to **detect** and **troubleshoot** fault.

5) Centralized management: The hub manages and controls the whole network.

Disadvantages:

1) Single point of failure: If the hub goes down, the whole network is dead.

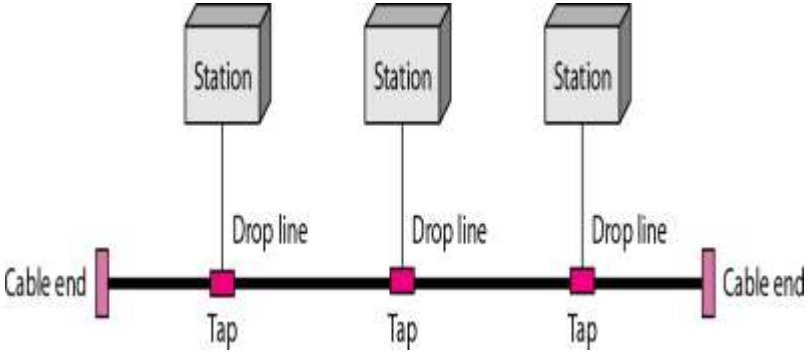
2) Cable length required is the more compared to bus/ring topologies.

3) Number of nodes in network depends on **capacity of hub**.

Example: Local area network

Bus Topology

- Multipoint connection
- All the devices are connected to the single cable called bus (backbone)
- Devices are connected to the bus by drop-lines and taps.
- A drop-line is a connection running between the device and the bus (main cable).
- A tap is a connector that links to the bus
- Limit on number of taps a bus can support and distance between those taps-(as signal travels along backbone some energy is transferred to heat which makes it weaker as it travels farther)



A bus topology connecting three stations

Bus Topology

Advantages:

1) Easy installation.
2) Cable required is the least compared to mesh/star topologies.
3) Redundancy is eliminated.
4) Costs less (Compared to mesh/star topologies).
5) Mostly used in small networks. Good for LAN.

Disadvantages:

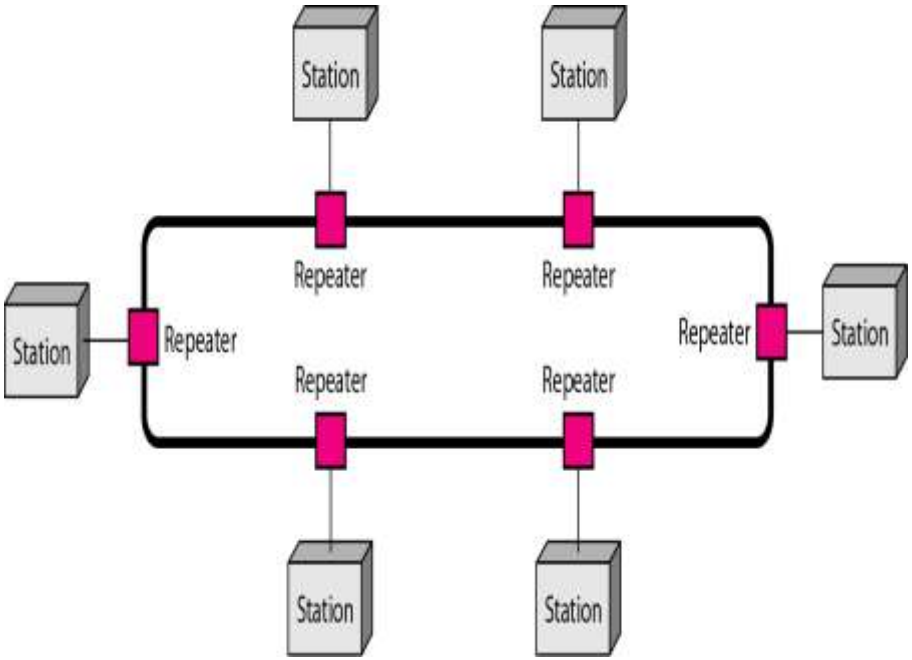
1) Difficult to detect and troubleshoot fault.
2) Signal reflection at the taps can cause degradation in quality.
3) A fault/break in the cable stops all transmission.
4) There is a limit on
i) Cable length
ii) Number of nodes that can be connected.
5) Security is very low because all the devices receive the data sent from the source.

Ring Topology

- Each device is connected to the next device, forming a ring
- There are only two neighbours for each device.
- Data travels around the network in one direction till the destination is reached.
- Each device has a repeater.
- A repeater

→ receives a signal on transmission-medium &

→ regenerates & passes the signal to next device.



A ring topology connecting six stations

Ring Topology

Advantages:

1) Easy installation and reconfiguration.

To add/delete a device, requires changing only 2 connections.

2) Fault isolation is simplified.

If one device does not receive a signal within a specified period, it can issue an alarm.

The alarm alerts the network-operator to the problem and its location.

3) Congestion reduced: Because all the traffic flows in one direction.

Disadvantages:

1) Unidirectional traffic.- makes it slower.

2) A fault in the ring/device stops all transmission.

The above 2 drawbacks can be overcome by using dual ring.

3) There is a limit on

i) Cable length &

ii) Number of nodes that can be connected.

4) Slower: Each data must pass through all the devices between source and destination.

Categories of Networks

Few criteria –size, geographical coverage and ownership to make this distinction.

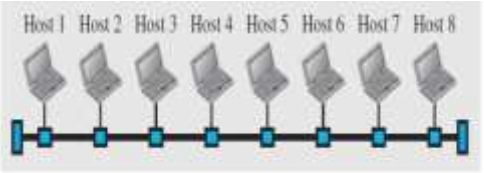
- **Local Area Networks (LANs)**
  - Short distances
  - Designed to provide local interconnectivity
- **Wide Area Networks (WANs)**
  - Long distances
  - Provide connectivity over large areas

Local Area Networks (LANs)

- It is used to connect computers in a single office, building or campus.
- Privately owned network.
- A LAN can be simple or complex.
  1) Simple: LAN may contain 2 PCs and a printer.
  2) Complex: LAN can extend throughout a company.(extend to audio and video devices.)
- Each host in a LAN has an address that uniquely defines the host in the LAN.
- A packet sent by a host to another host carries both source host‟s and destination host‟s addresses.

Local Area Networks (LANs) (cont…)

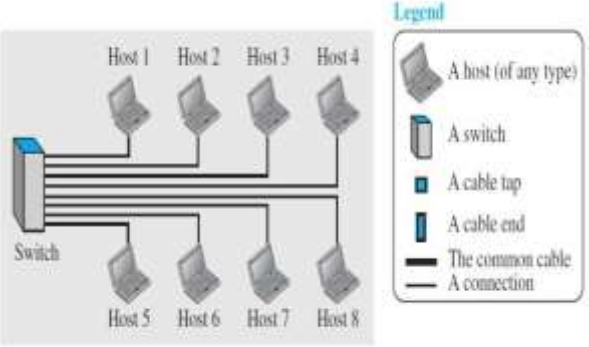
- LAN with common cable : packet sent by host is received by all hosts. Intended host keep the packet other drop the packet.



a. LAN with a common cable (past)

- LAN use a smart connecting switch.

• The switch is able to
→ recognize the destination address of the packet & guide the packet to its destination.
→ reduces the traffic in the LAN
→ allows more than one pair to communicate with each other at the same time.



b. LAN with a switch (today)

Advantages:
Resource Sharing: Computer resources like printers and hard disks can be shared by all devices on the network.

Wide Area Networks (WAN)

- WAN can cover wider geographical area. It can cover cities, states, countries and even world.
- WAN interconnects connecting devices such as switches, routers, or modems.
- Normally, WAN is

→ created & run by communication companies (Ex: BSNL, Airtel)

→ leased by an organization that uses it.

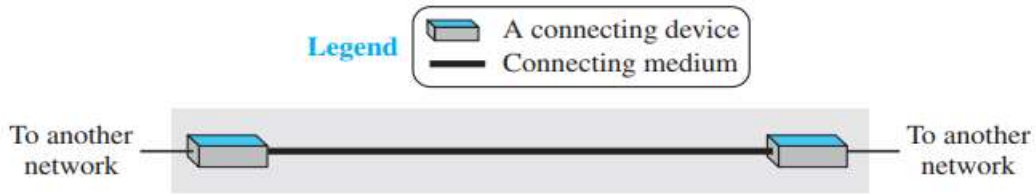
Two distinct examples of WANs:

- Point-to-point WAN
- Switched WAN

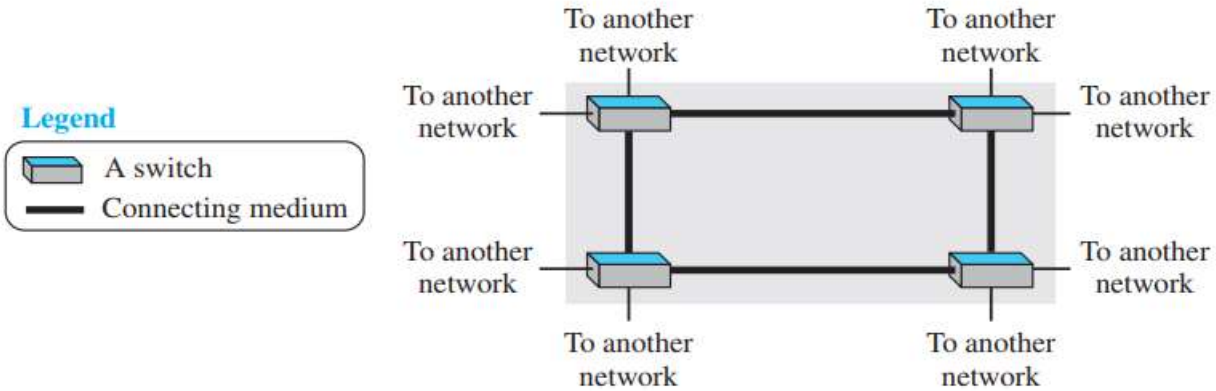
Wide Area Networks (WAN) (cont…)

Two distinct examples of WANs:

- Point-to-point WAN:A point-to-point WAN is a network that connects 2 communicating devices through a transmission media



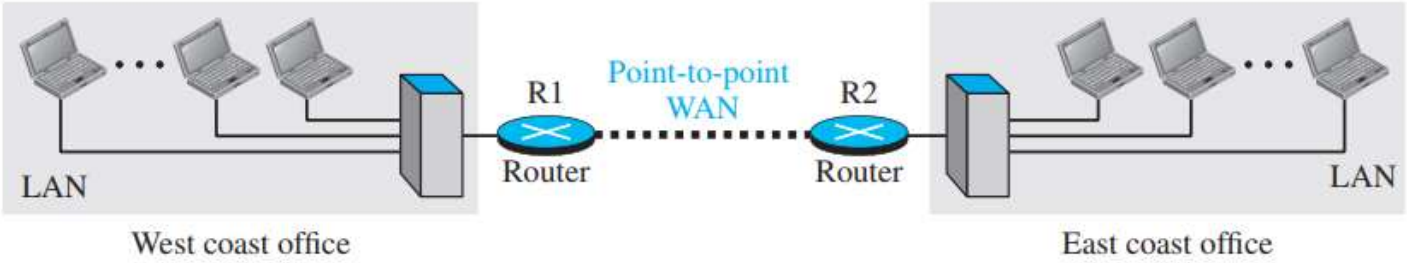
- Switched WAN: A switched WAN is a network with more than two ends.
  - The switched WAN can be the backbones that connect the Internet.
  - A switched WAN is a combination of several point-to-point WANs that are connected by switches



Wide Area Networks (WAN) (cont…)

Internetwork:

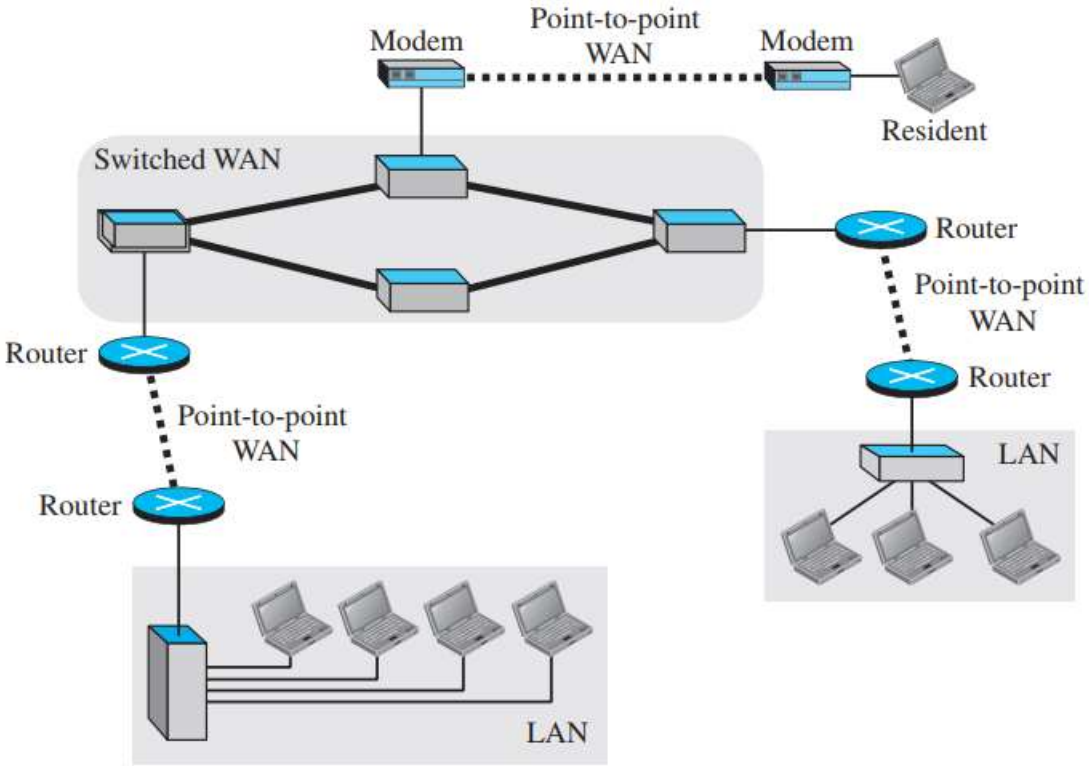
- A network of networks is called an **internet**. (inter-network)
- EX: Assume that an organization has two offices, First office is on the east coast & Second office is on the west coast.
- Each office has a LAN that allows all employees in the office to communicate with each other.
- To allow communication between employees at different offices, the management leases a point-to-point dedicated WAN from a ISP and connects the two LANs.
- When a host in the west coast office sends a message to another host in the same office, the router blocks the message, but the switch directs the message to the destination.
- On the other hand, when a host on the west coast sends a message to a host on the east coast, router R1 routes the packet to router R2, and the packet reaches the destination.



Wide Area Networks (WAN) (cont…)

Internetwork:

Internet with several LANs and WANs connected. One of the WANs is a switched WAN with four switches.



Switching

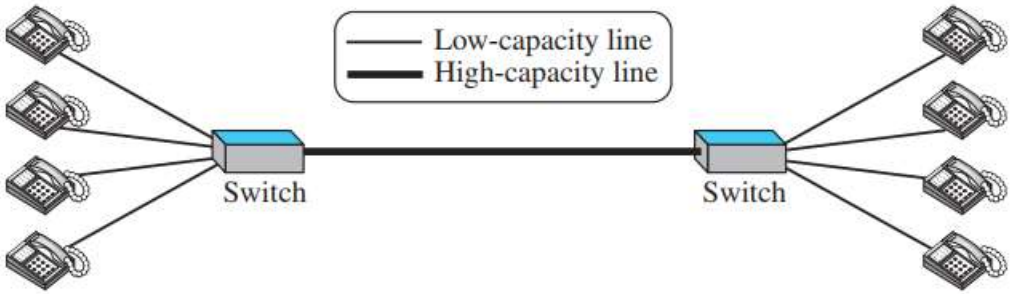
Switching:

- An internet is a switched network in which a **switch** connects at **least two links together.**

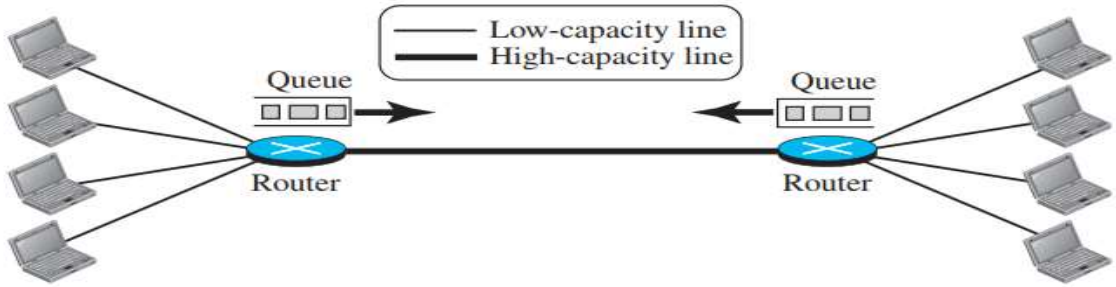
- A switch needs to **forward data** from a network to another network when required.

- **Two types** of switched networks are
  1) circuit-switched and
  2) packet-switched networks.

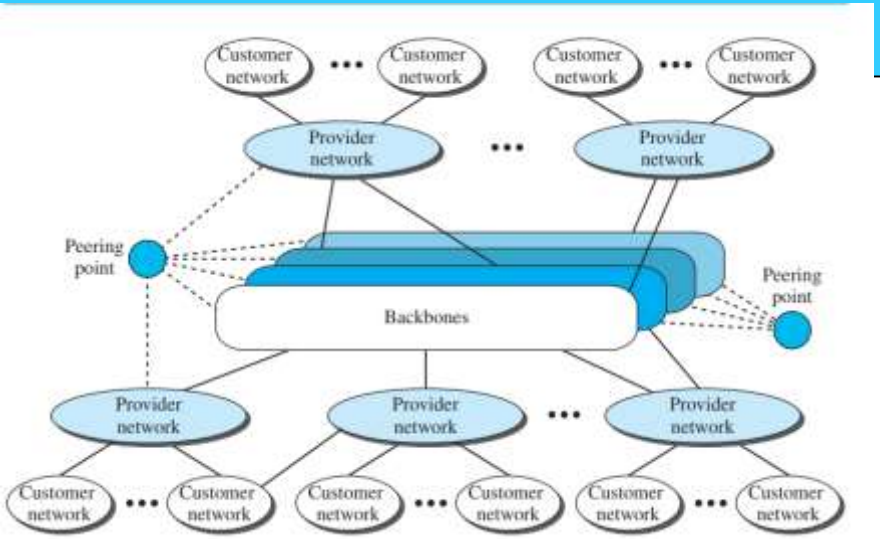
- **Circuit switched network:**
  - A dedicated connection, called a **circuit**, is always available between the two end systems. The switch can only make it **active or inactive**.
  - Ex: In the figure below the 4 telephones at each side are connected to a switch.
  - The switch connects a telephone at one side to a telephone at the other side.
  - A high-capacity line can handle 4 voice communications at the same time.
  - The capacity of high line can be shared between all pairs of telephones.
  - The switch is used for only forwarding.
  - **Advantage:**
    - A circuit-switched network is **efficient** only when it is working at its full capacity.
  - **Disadvantage:**
    - Most of the time, the network is **inefficient** because it is working at partial capacity.



- **Packet switched network:**
  - In a computer network, the communication between the 2 ends is done in blocks of data called **packets**.
  - The switch is used for both **storing** and **forwarding** because a packet is an independent entity that can be stored and sent later..
  - As shown in Figure below, the 4 computers at each side are connected to a router.
  - A router has a **queue** that can store and forward the packet.
  - The high-capacity line has twice the capacity of the low-capacity line.
  - If only 2 computers (one at each site) need to communicate with each other, there is no waiting for the packets.
  - However, if packets arrive at one router when high-capacity line is at its full capacity, the packets should be stored and forwarded.
  - **Advantages:**
  - A packet-switched network is more efficient than a circuit switched network.
  - **Disadvantage:**
  - The packets may encounter some delays.



THE INTERNET



- The most notable internet is called the Internet (uppercase I) and is composed of thousands of interconnected networks.

• Internet is made up of

1) Backbones
2) Provider networks &
3) Customer networks
4) Backbones
    - Backbones are large networks owned by communication companies such as BSNL and Airtel.
    - The backbone networks are connected through switching systems, called peering points

5) Provider Networks (smaller network)
    - Provider networks are connected to backbones and sometimes to other provider networks.

3) Customer Networks
    - Customer networks actually use the services provided by the Internet.
    - Customer networks pay fees to provider networks for receiving services.

• Backbones and provider networks are also called Internet Service Providers (ISPs).

• The provider networks are often referred to as national or regional ISPs.

Accessing the Internet

- The Internet today is an internetwork that allows any user to become part of it.
- However, the user needs to be physically connected to an ISP.
- The physical connection is normally done through a point-to-point WAN.

1) Using Telephone Networks

Most homes and small businesses already have telephone service, which can also be used to connect to the Internet in two ways:

Dial-up service – Uses a modem to connect to the ISP through a normal phone line. It is very slow, and the line cannot be used for voice calls while connected. Suitable only for small residences.

DSL (Digital Subscriber Line) service – A faster upgrade by telephone companies that allows the same line to carry both Internet and voice calls at the same time, making it more efficient for homes and small businesses.

Accessing the Internet (cont..)

2) Using Cable Networks

- A residence can be connected to the Internet by using cable service.
- Cable service provides a higher speed connection.
- The speed varies depending on the number of neighbours that use the same cable.

Example: A family uses their cable TV provider's Internet service for streaming and browsing.

3) Using Wireless Networks

- A residence can use a combination of wireless and wired connections to access the Internet.
- A residence can be connected to the Internet through a wireless WAN.

Example: A small business connects to the Internet via Wi-Fi from a wireless broadband provider.

4) Direct Connection to the Internet

- A large organization can itself become a local ISP and be connected to the Internet.
- The organization
  → leases a high-speed WAN from a carrier provider and connects itself to a regional ISP.
- For example, a large university with several campuses can create an internetwork and then conn

Internet History

- Internet has evolved from a private network to a global one in less than 40 years.

Early History

- Before 1960 there were some communication networks such as telegraph and telephone networks.
- Suitable for constant-rate communication (after a connection was made between two users, the encoded message(telegraphy) or voice (telephony) could be exchanged).
- Computer network should be able to handle bursty data (data received at variable rates at different times)

Birth of packet switched networks:

- The theory of packet switching for bursty traffic was first presented by Leonard Kleinrock in 1961 at MIT.
- At the same time, two other researchers, Paul Baran at Rand Institute and Donald Davies at National Physical Laboratory in England, published some papers about packet-switched networks.

Internet History (cont…)

ARPANET:

- In the mid-1960s, mainframe computers in research organizations were stand-alone devices.
- Computers from different manufacturers were unable to communicate with one another.
- The Advanced Research Projects Agency (ARPA) in the Department of Defence (DOD) was interested in finding a way to connect computers so that the researchers share their findings, thereby reducing costs and eliminating duplication of effort.
- In 1967, at an Association for Computing Machinery (ACM) meeting, ARPA presented its ideas for the Advanced Research Projects Agency Network (ARPANET), a small network of connected computers.
- The idea was that each host computer (not necessarily from the same manufacturer) would be attached to a specialized computer, called an *interface message processor* (IMP).
- The IMPs, in turn, would be connected to each other. Each IMP had to be able to communicate with other IMPs as well as with its own attached host.
- By 1969, ARPANET was a reality.
- Four nodes, at the University of California at Los Angeles (UCLA), the University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and the University of Utah, were connected via the IMPs to form a network.
- Software called the *Network Control Protocol* (NCP) provided communication between the hosts.

Birth of the Internet

- In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the *Internetting Project*.
- They wanted to link dissimilar networks so that a host on one network could communicate with a host on another.
- There were many problems to overcome: diverse packet sizes, diverse interfaces, and diverse transmission rates, as well as differing reliability requirements.
- Cerf and Kahn devised the idea of a device called a *gateway* to serve as the intermediary hardware to transfer data from one network to another.

TCP/IP

- New version of NCP (Network Control Protocol) - transmission control protocol (TCP) included concepts such as encapsulation, the datagram, and the functions of a gateway.

- Authorities made a decision to split TCP into two protocols: Transmission Control Protocol (TCP) and Internet Protocol (IP). IP would handle datagram routing while TCP would be responsible for higher level functions such as segmentation, reassembly, and error detection. The new combination became known as TCP/IP.

Internet History (cont…)

MILNET:
In 1983, ARPANET split into two networks: Military Network (MILNET) for military users and ARPANET for non-military users.

CSNET:

- Creation of CSNET in 1981.
- Computer Science Network (CSNET) was a network sponsored by the National Science Foundation (NSF).
- The network was conceived by universities that were ineligible to join ARPANET due to an absence of ties to the Department of Defense.
- CSNET was a
  - Less expensive network;
  - there were no redundant links and
  - the transmission rate was slower.
- By the mid-1980s, most U.S. universities with computer science departments were part of CSNET.

Internet History (cont…)

NSFNET

- With the success of CSNET, the NSF in 1986 sponsored the National Science Foundation Network (NSFNET), a backbone that connected five supercomputer centres located throughout the United States.
- In 1990, ARPANET was officially retired and replaced by NSFNET.

ANSNET

- In 1991, the U.S. government decided that NSFNET was not capable of supporting the rapidly increasing Internet traffic.
- Three companies, IBM, Merit, and Verizon, filled the void by forming a non-profit organization called Advanced Network & Services (ANS) to build a new, high-speed Internet backbone called Advanced Network Services Network (ANSNET).

Internet Today

World Wide Web

- The 1990s saw the explosion of Internet applications due to the emergence of the World Wide Web (WWW).

Multimedia

Recent developments in the multimedia applications such as

- voice over IP (telephony),
- video over IP (Skype),
- view sharing (YouTube), and
- television over IP (PPLive)

increased the number of users and the amount of time each user spends on the network.

Peer-to-Peer Applications (P2P)

Peer-to-peer networking is also a new area of communication

Ex: BitTorrent (file sharing).

Internet Standards

- An Internet standard is an official rule that everyone working with the Internet must follow.
- It starts as an Internet Draft (a work in progress, valid for 6 months).
- If approved, it becomes a Request for Comment (RFC), which is numbered and shared publicly. RFCs then move through different maturity levels until they become a full Internet standard.

Maturity Levels

• An RFC, during its lifetime, falls into one of 6 maturity levels

1) Proposed Standard

Proposed standard is specification that is stable, well-understood & of interest to Internet community.

Specification is usually tested and implemented by several

2) Draft Standard

• A proposed standard is elevated to draft standard status af independent (i.e., two different groups create the same protocol correctly).and interoperable implementations (they can talk issues).

3) Internet Standard

A draft standard reaches Internet standard status after demo successful implementation.

4) Historic

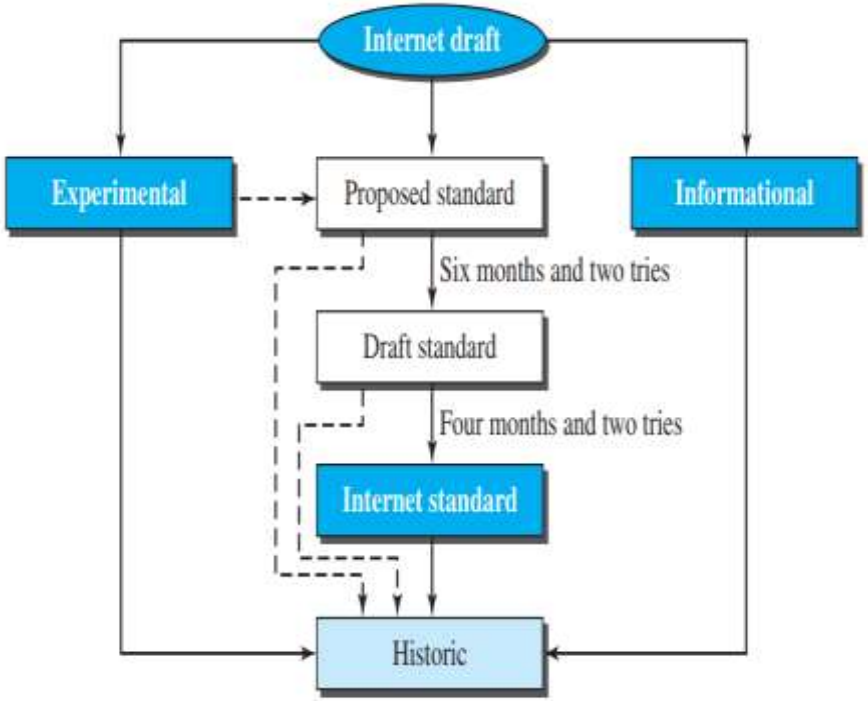
The historic RFCs are significant from a historical persp They either

→ have been replaced by later specifications or

→ have never passed the necessary maturity levels to become

5) Experimental

An RFC classified as experimental describes work relate situation. Such an RFC should not be implemented in any service.



6) Informational

An RFC classified as informational contains general, historical, or tutorial information related to the Internet.

Requirement Levels

•RFCs are classified into 5 requirement levels:

1) Required

- An RFC labelled required must be implemented by all Internet systems to ensure they can work together.
- For example, IP and ICMP (Internet Control Message Protocol) are required protocols.

2) Recommended

- An RFC labeled recommended is not required for minimum conformance.
- It is recommended because of its usefulness.
- For example, FTP is recommended protocols.

3) Elective

- An RFC labeled elective is not required and not recommended.
- However, a system can use it for its own benefit.

4) Limited Use

- An RFC labeled limited use should be used only in limited situations.
- Most of the experimental RFCs fall under this category.

5) Not Recommended

- An RFC labeled not recommended is inappropriate for general use.
- Normally a historic RFC may fall under this category.

Internet Administration

1) ISOC

• ISOC is a non-profit organization formed to provide support for Internet standards process

• ISOC maintains and supports other Internet administrative bodies such as IAB, IETF, IRTF, and IANA.

2) IAB

• IAB is the technical advisor to the ISOC.

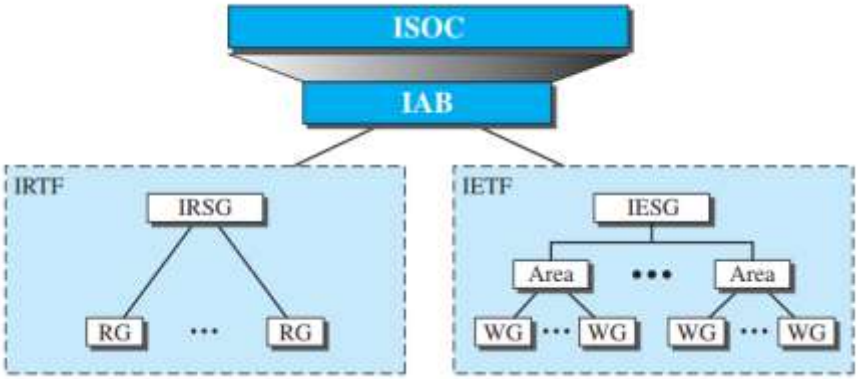
• Two main purposes of IAB:

i) To oversee the continuing development of the TCP/IP Protocol Suite

ii) To serve in a technical advisory to research members of the Internet community.

• Another responsibility of the IAB is the editorial management of the RFCs.

• IAB has 2 primary components: i) IETF and ii) IRTF.



Standards and Administration (cont..)

i) IETF (Internet Engineering Task Force)

- IETF is responsible for identifying operational problems & proposing solutions to the problems
- IETF also develops and reviews specifications intended as Internet standards.
- The working groups are collected into areas, and each area concentrates on a specific topic.
- Currently 9 areas have been defined. The areas include applications, protocols, routing, network management next generation and security.

ii) IRTF

- IRTF focuses on long-term research topics related to Internet protocols, applications, architecture, and technology.

Chapter 2

Network Models

PROTOCOL LAYERING

- A protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.
- When communication is
  - Simple -only one simple protocol.
  - complex, we need to divide the task b/w different layers. We need a protocol at each layer, or protocol layering.

Scenarios

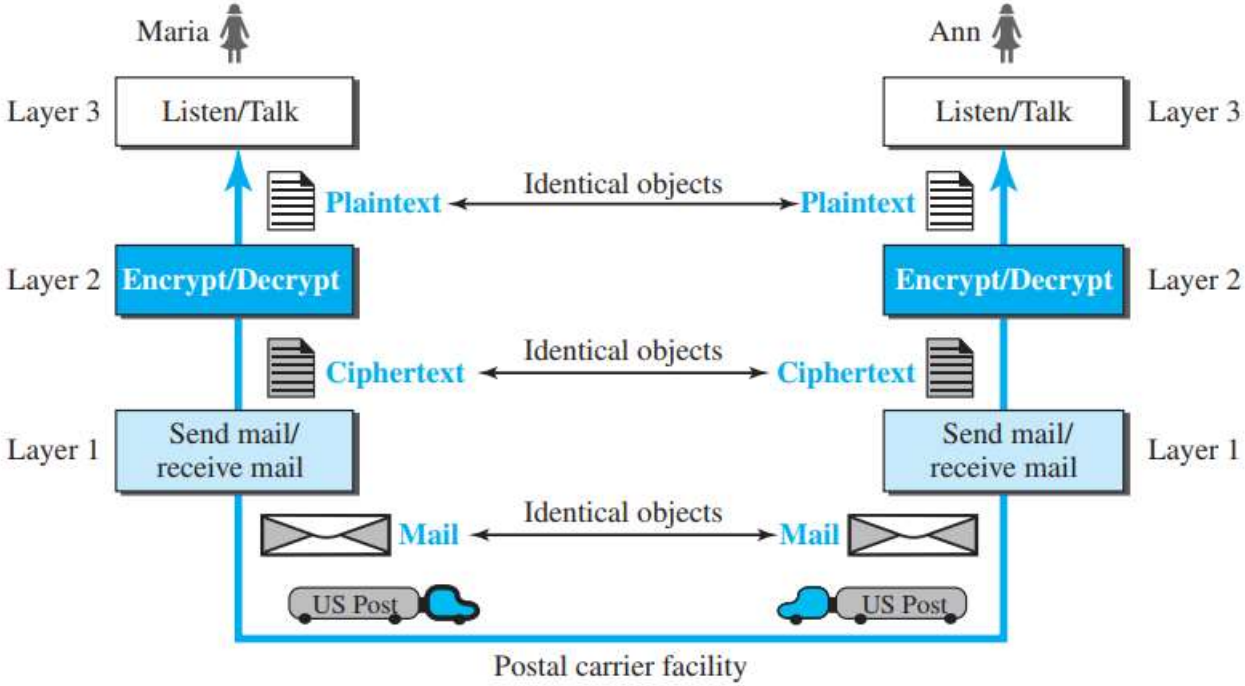
First Scenario

- Communication is so simple that it can occur in only one layer.
- Assume Maria and Ann are neighbours with a lot of common ideas.
- Communication between Maria and Ann takes place in one layer, face to face, in the same language

PROTOCOL LAYERING (Cont..)

Second Scenario

- Maria and Ann communicate using regular mail through the post office .
- However, they do not want their ideas to be revealed by other people if the letters are intercepted.
- They agree on an encryption/decryption technique.
- The sender of the letter encrypts it to make it unreadable by an intruder; the receiver of the letter decrypts it to get the original letter.



PROTOCOL LAYERING (Cont..)

- Protocol layering enables us to divide a complex task into several smaller and simpler tasks.
- However, if Maria and Ann decide that the encryption/decryption done by the machine is not enough to protect their secrecy, they would have to change the whole machine. In the present situation, they need to change only the second layer machine; the other two can remain the same. This is referred to as modularity
- Modularity means independent layers.
- A layer (module) can be defined as a black box with inputs and outputs, without concern about how inputs are changed to outputs.
- If two machines provide the same outputs when given the same inputs, they can replace each other.
- Protocol layering enables us to divide a complex task into several smaller and simpler tasks.
- Advantages:
    1) It allows us to separate the services from the implementation.-A layer needs to be able to receive a set of services from the lower layer and to give the services to the upper layer; we don't care about how the layer is implemented.
    2) There are intermediate systems that need only some layers, but not all layers.
- Disadvantage:
    1) Having a single layer makes the job easier. There is no need for each layer to provide a service t

Principles of Protocol Layering

1) First Principle

• If we want bidirectional communication, we need to make each layer able to perform 2 opposite tasks, one in each direction.

• For example, the third layer task is to listen (in one direction) and talk (in the other direction)

.

2) Second Principle

• The two objects under each layer at both sites should be identical.

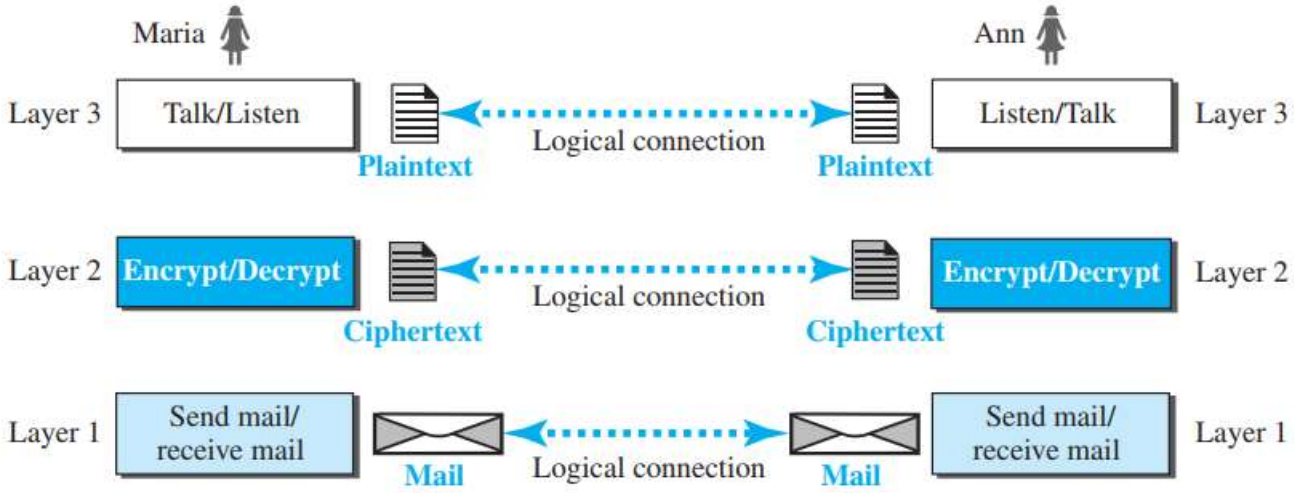
• For example, the object under layer 3 at both sites should be a plaintext letter.

Principles of Protocol Layering

Logical Connections

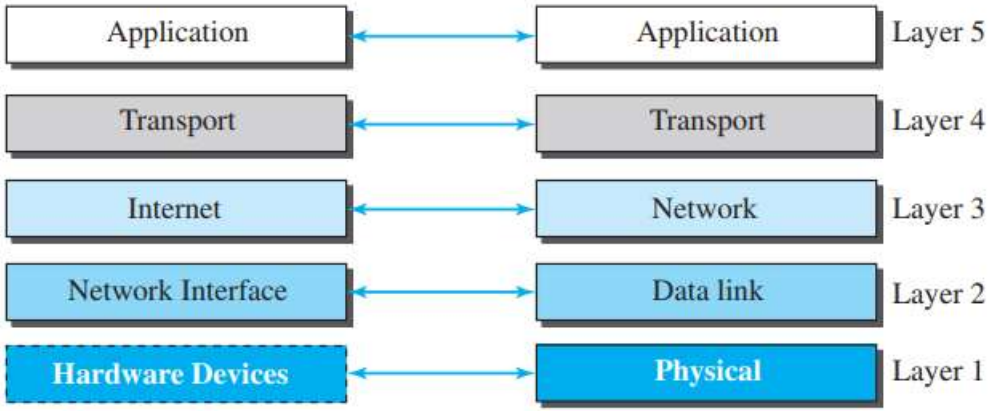
• There is a logical (imaginary) connection at each layer through which two end systems can send the object created from that layer.

•Each layer believes it is directly communicating with the same layer on the other side.
•But in reality, communication happens through the services of lower layers.



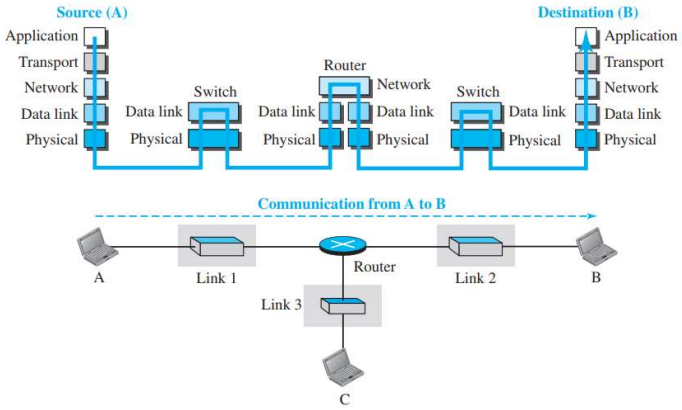
TCP/IP PROTOCOL SUITE

- TCP/IP is a protocol-suite used in the Internet today.
- A protocol suite means a collection of protocols that work together to provide complete communication over a network.
- The term hierarchical means that each upper level protocol is supported by the services provided by one or more lower level protocols.
- TCP/IP is thought of as a five-layer model.



TCP/IP PROTOCOL SUITE (Cont..)

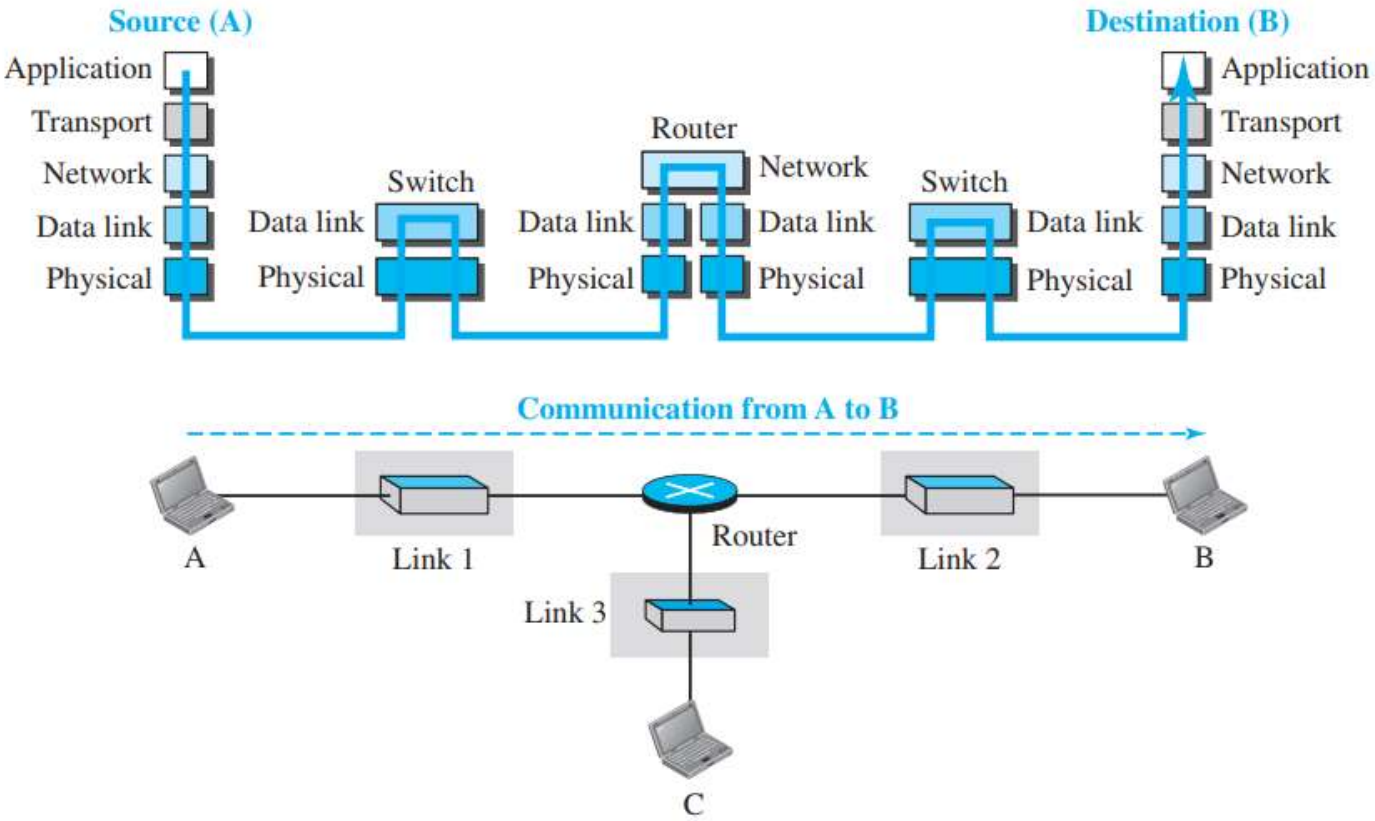
Layered Architecture



Let us assume that computer A communicates with computer B

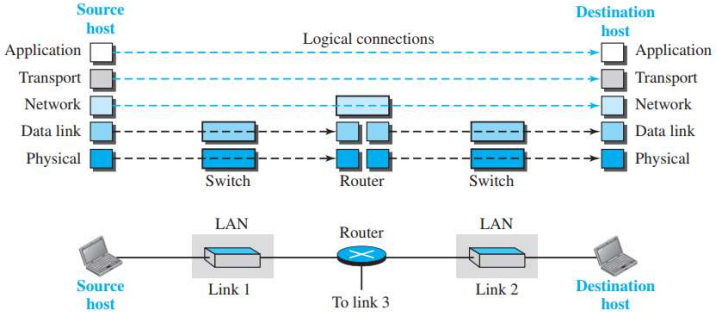
- As shown in the Figure, we have five communicating device s:
  1) Source host(computer A)
  2) Link-layer switch in link 1
  3) Router
  4) Link-layer switch in link 2
  5) Destination host (computer B).
- Each device is involved with a set of layers depending on the role of the device in the internet.
- The two hosts are involved in all five layers.
- The source host
  → creates a message in the application layer and
  → sends the message down the layers so that it is physically sent to the destination host.
- The destination host
  → receives the message at the physical layer and
  → then deliver the message through the other layers to the application layer.
- The router is involved in only three layers; there is no transport or application layer.
- A link-layer switch is involved only in two layers: i) data-link and ii) physical.

TCP/IP PROTOCOL SUITE (Cont..)

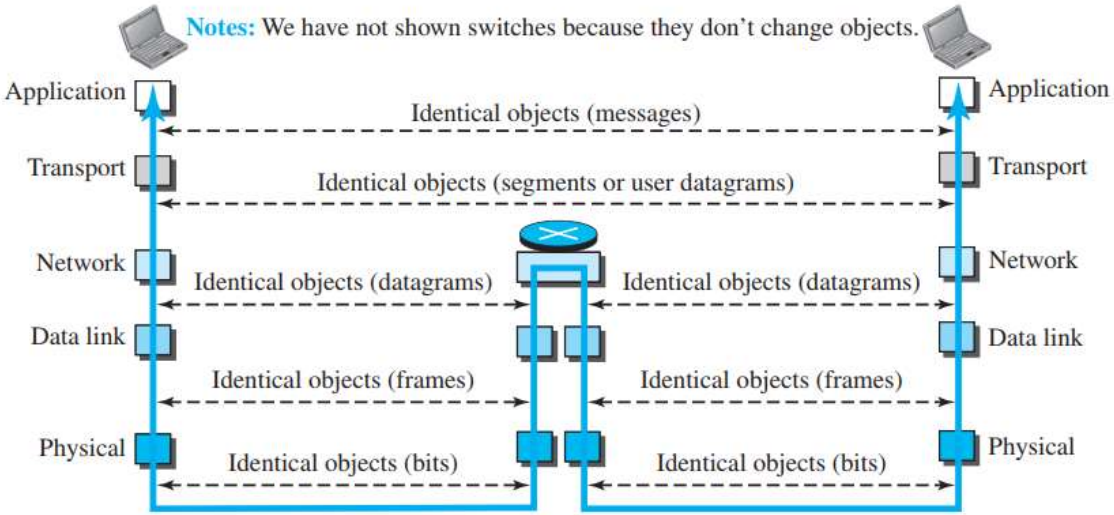


TCP/IP PROTOCOL SUITE (Cont..)

Layers in TCP/IP Protocol suite



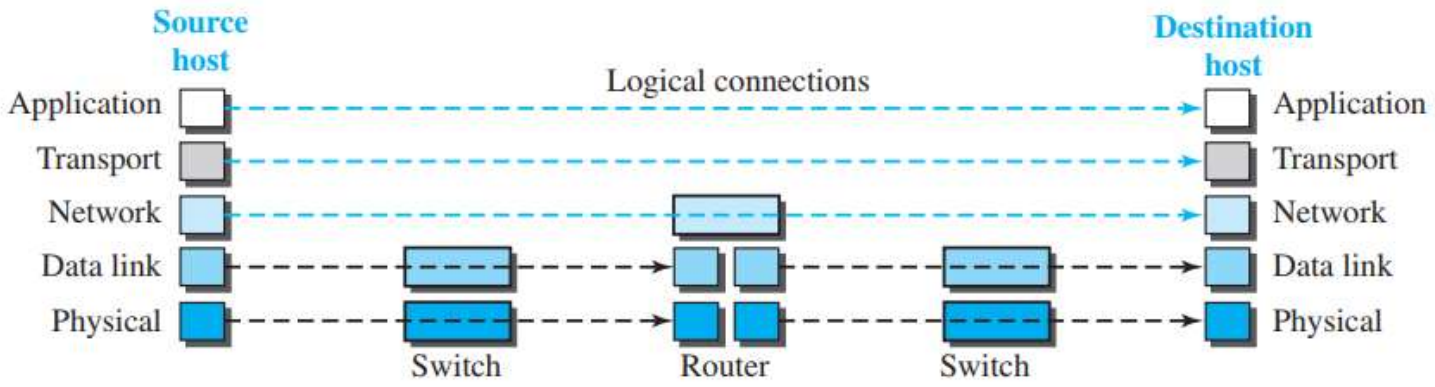
- The duty of the
  - Application, Transport, and Network layers is end-to-end.
  - Data-link and Physical layers is hop-to-hop (hop is a host or router).
- The domain of duty of the
  - top three layers is the internet. (overall communication across networks).
  - two lower layers is the link. (local communication between directly connected devices).

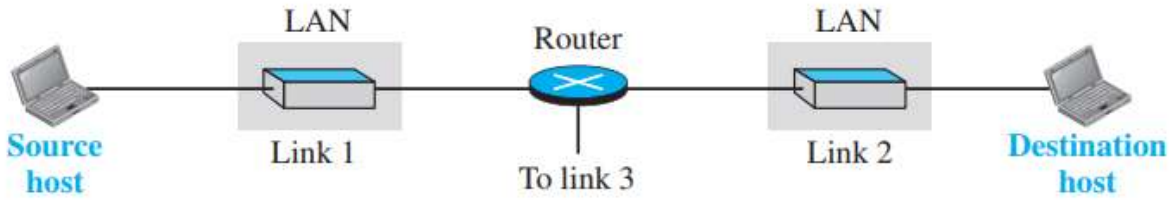


TCP/IP PROTOCOL SUITE (Cont..)

Layers in TCP/IP Protocol suite

- The duty of the
  - Application, Transport, and Network layers is end-to-end.
  - Data-link and Physical layers is hop-to-hop (hop is a host or router).
- The domain of duty of the
  - top three layers is the internet. (overall communication across networks).
  - two lower layers is the link. (local communication between directly connected devices).

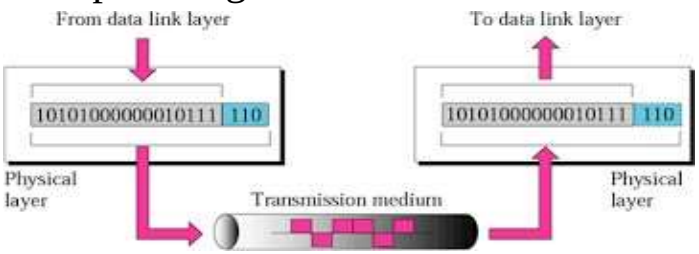




Description of Each layer

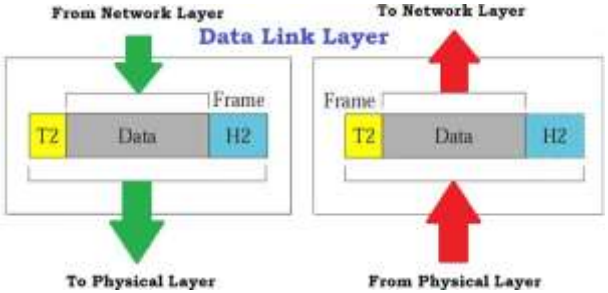
Physical Layer

- The physical layer is responsible for movements of individual bits from one node to another node.
- Transmission media is another hidden layer under the physical layer.
- Two devices are connected by a transmission medium (cable or air).
- The transmission medium does not carry bits; it carries electrical or optical signals.
- The physical layer
  - → receives bits from the data-link layer &
  - → sends through the transmission media.



Data Link Layer

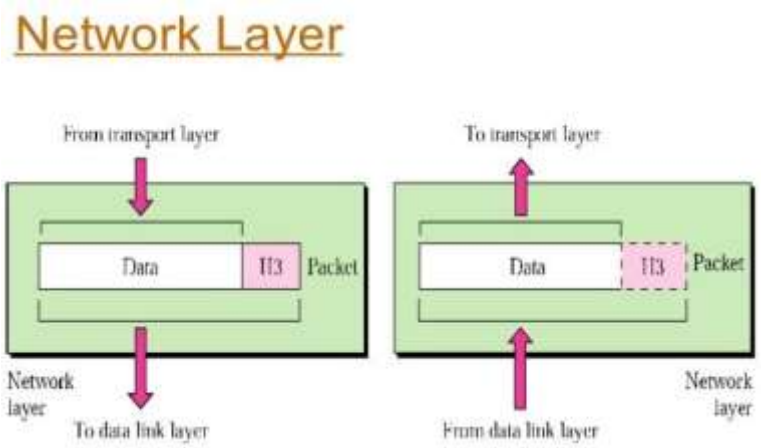
- Data-link-layer (DLL) is responsible for moving frames from one node to another node over a link.
- The link can be wired LAN/WAN or wireless LAN/WAN.
- The data-link layer
  - → gets the datagram from network layer
  - → encapsulates the datagram in a packet called a frame.
  - → sends the frame to physical layer.
- Some protocols provide complete error detection and correction; some protocols provide only Error correction



Description of Each layer

Network Layer

- The network layer is responsible for source-to-destination transmission of data.
- The network layer is also responsible for routing the packet.
- The routers choose the best route for each packet.
- Why we need the separate network layer?
  1) The separation of different tasks between different layers.
  2) The routers do not need the application and transport layers.
- TCP/IP model defines 4 protocols:
  1) ARP (Address Resolution Protocol)
  2) ICMP (Internet Control Message Protocol)
  3) IGMP (Internet Group Message Protocol)
  4) DHCP (Dynamic Host Configuration Protocol)



1) IP

IP is the main protocol of the network layer.

IP defines the format and the structure of addresses.

IP is also responsible for routing a packet from its source to its destination.

It is a connection-less & unreliable protocol.

i) Unreliable protocol means

→ IP does not make any guarantee about delivery of the data.

→ Packets may get dropped during transmission.

It provides a best-effort delivery service.

Best effort means IP does its best to get the packet to its destination, but with no guarantees.

IP does not provide following services

→ flow control

→ error control

→ congestion control services.

If an application requires above services, the application should rely only on the transport layer protocol.

1) ARP (Address Resolution Protocol)

ARP is used to find the physical-address of the node when its Internet-address is known.
To find the MAC address when only the IP address is known.
Example: To send data inside a LAN, a computer must know the MAC address of the destination → ARP provides it.

2) IGMP (Internet Group Management Protocol)

To manage multicast groups.
Example: If you're streaming live video to many users, IGMP helps computers join or leave the group that receives the stream.

3) ICMP (Internet Control Message Protocol)

To report errors and network problems.
Example: If a router cannot deliver a packet (host unreachable, time expired), it sends an ICMP message back.

4) DHCP (Dynamic Host Configuration Protocol)

To automatically assign IP addresses to devices.
Example: When you connect your phone/laptop to Wi-Fi, you don't type the IP address manually DHCP gives it automatically.

Transport Layer

TL protocols are responsible for delivery of a message from a process to another process.

• The transport layer

→ gets the message from the application layer

→ encapsulates the message in a packet called a segment and

→ sends the segment to network layer.

• TCP/IP model defines 3 protocols: (as listed below)

1) TCP (Transmission Control Protocol)

TCP is a reliable connection-oriented protocol.

A connection is established b/w the sender and receiver before the data can be transmitted.

TCP provides

→ flow control

→ error control and

→ congestion control

2) UDP (User Datagram Protocol)

UDP is the simplest of the 3 transport protocols. It is an unreliable, connectionless protocol.

It does not provide flow, error, or congestion control. Each datagram is transported separately & independently It is suitable for application program that

→ Best for short or real-time data where speed matters more than reliability.

3) SCTP (Stream Control Transmission Protocol)

SCTP provides support for newer applications such as voice over the Internet.

It combines the best features of TCP and UDP

Application Layer

- The two application layers exchange messages between each other.
- Communication at the application layer is between two processes (two programs running at this layer).
- To communicate, a process sends a request to the other process and receives a response.
- Process-to-process communication is the duty of the application layer.

- TCP/IP model defines following protocols:

HTTP (HyperText Transfer Protocol) → Used for browsing websites (WWW).

SMTP (Simple Mail Transfer Protocol) → Used for sending emails.

FTP (File Transfer Protocol) → Used to transfer files between computers.

TELNET / SSH (TELecommunication NETwork/Secure Shell)→ Used for remote login (to control another computer).

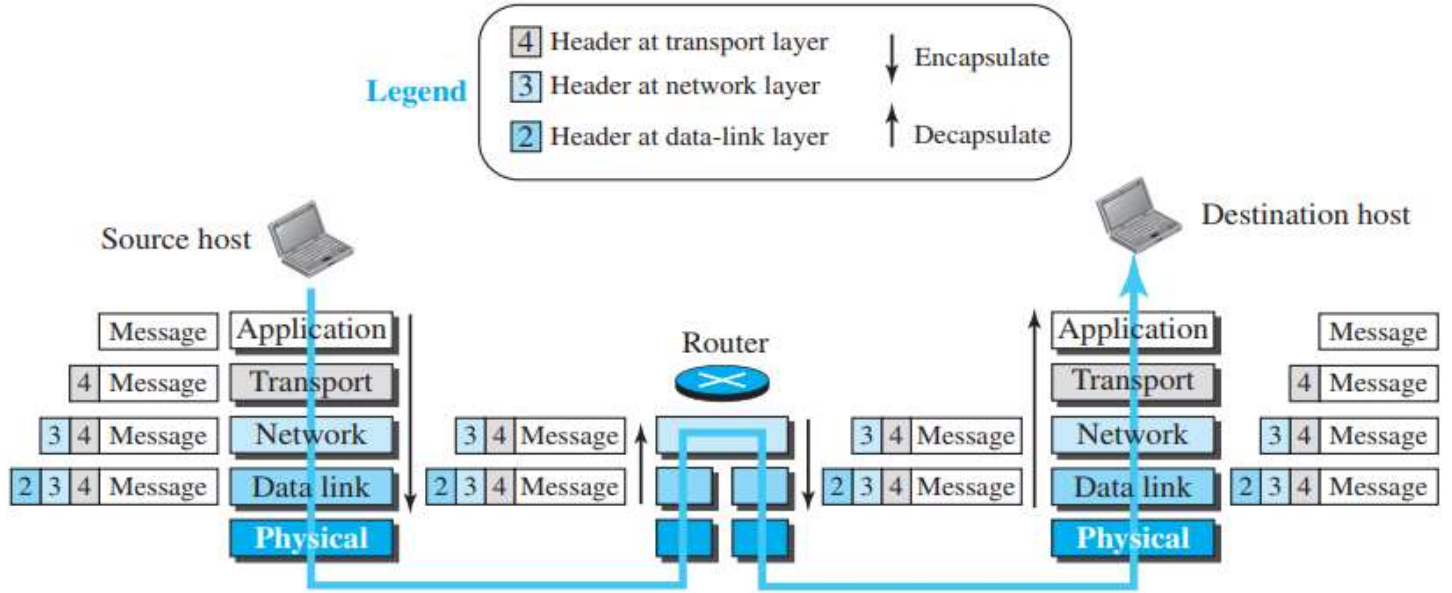
SNMP (Simple Network Management Protocol) → Used by network admins to manage and monitor devices (like routers, switches).

DNS (Domain Name System) → Converts website names (like google.com) into IP addresses.

IGMP (Internet Group Management Protocol) → Helps manage groups for multicasting (sending data to many users at once, e.g., live streaming).

Encapsulation and Decapsulation

- One of the important concepts in protocol layering in the Internet.
- Figure below shows, encapsulation in the source host, decapsulation in the destination host, and encapsulation and decapsulation in the router.



Encapsulation at the Source Host

At the source, we have only encapsulation

1) At the application layer, the data to be exchanged is referred to as a message.

A message normally does not contain any header .

The message is passed to the transport layer.

2) The transport layer takes the message as the payload.

TL adds its own header to the payload.

The header contains

→ identifiers of the source and destination application programs

→ information needed for flow, error control, or congestion control.

The segment is passed to the network layer.

3) The network layer takes the transport-layer packet as payload.

NL adds its own header to the payload. The header contains

→ addresses of the source and destination hosts

→ some information used for error checking of the header &

→ fragmentation information.

The network-layer packet is called a datagram. The datagram is passed to the data-link layer.

4) The data-link layer takes the network-layer packet as payload.

DLL adds its own header to the payload.

The header contains the physical addresses of the host or the next hop (the router). The link-layer packet is called a frame.

The frame is passed to the physical layer for transmission

1. After the set of bits are delivered to the data-link layer, this layer decapsulates the datagram from the frame and passes it to the network layer.

2. The network layer only inspects the source and destination addresses in the Datagram header and consults its forwarding table to find the next hop to which the datagram is to be delivered. The datagram is then passed to the data-link layer of the next link.

3. The data-link layer of the next link encapsulates the datagram in a frame and passes it to the physical layer for transmission.

Decapsulation at the Destination Host

• At the destination host, each layer

→ decapsulates the packet received from lower layer

→ removes the payload and delivers the payload to the next-higher layer protocol until the message reaches the application layer

Addressing

Addressing in Networking

For any communication, we always need **two addresses**:

•**Source address** → who is sending

•**Destination address** → who should receive

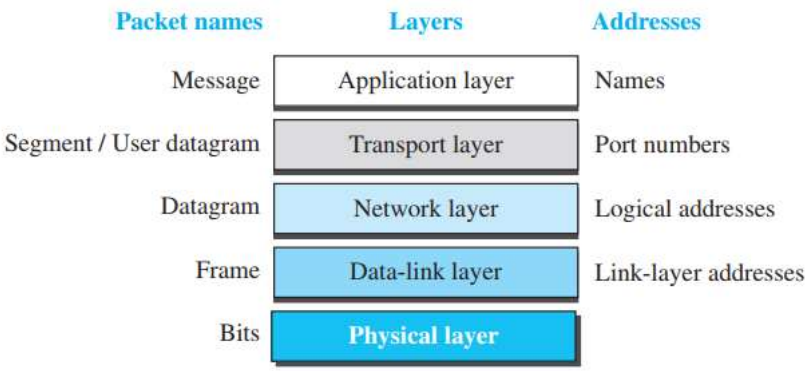
In the Internet model, different **layers** use different types of addresses:

1. Application Layer (Names / IDs)

•Uses names (like website names or email IDs).

•Example:

- www.google.com (website name)
- someone@gmail.com (email address)



2. Transport Layer (Port Numbers)

•Uses port numbers to identify programs running on a computer.

•Example:

- Web browser → Port 80 (HTTP) or 443 (HTTPS)

3. Network Layer (IP Address)

•Uses IP addresses to identify devices globally across the Internet.

4. Data-Link Layer (MAC Address)

•Uses MAC (Media Access Control) addresses

5. **No Address at Physical Layer**

•Physical layer just deals with bits (0s and 1s).

•Bits don’t need addresses.

Multiplexing and Demultiplexing

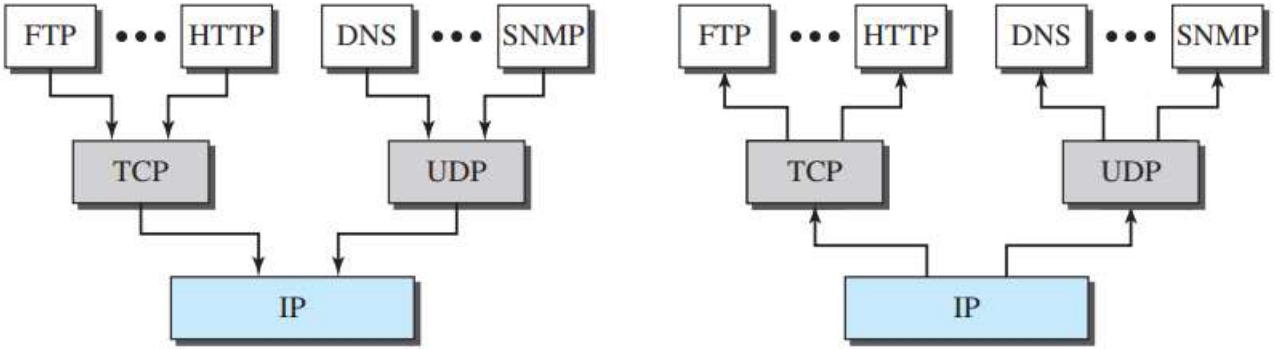
• Multiplexing means a protocol at a layer can encapsulate a packet from several next-higher Layer protocols (one at a time)

Demultiplexing means a protocol can decapsulate and deliver a packet to several next-higher layer protocols (one at a time).

1) At transport layer, either UDP or TCP can accept a message from several application-layer protocols.

2) At network layer, IP can accept Datagram from UDP

3) At data-link layer, a frame may carry the payload coming from IP .



a. Multiplexing at source

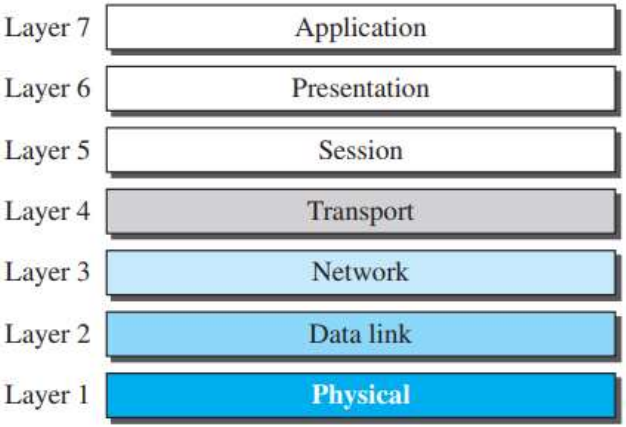
b. Demultiplexing at destination

OSI MODEL

- OSI model was developed by ISO.
- Purpose: OSI was developed to allow systems with different hardware, software or operating system to communicate with each other.
- OSI is a network-model that defines the protocols for network communications.
- OSI has 7 layers as follows

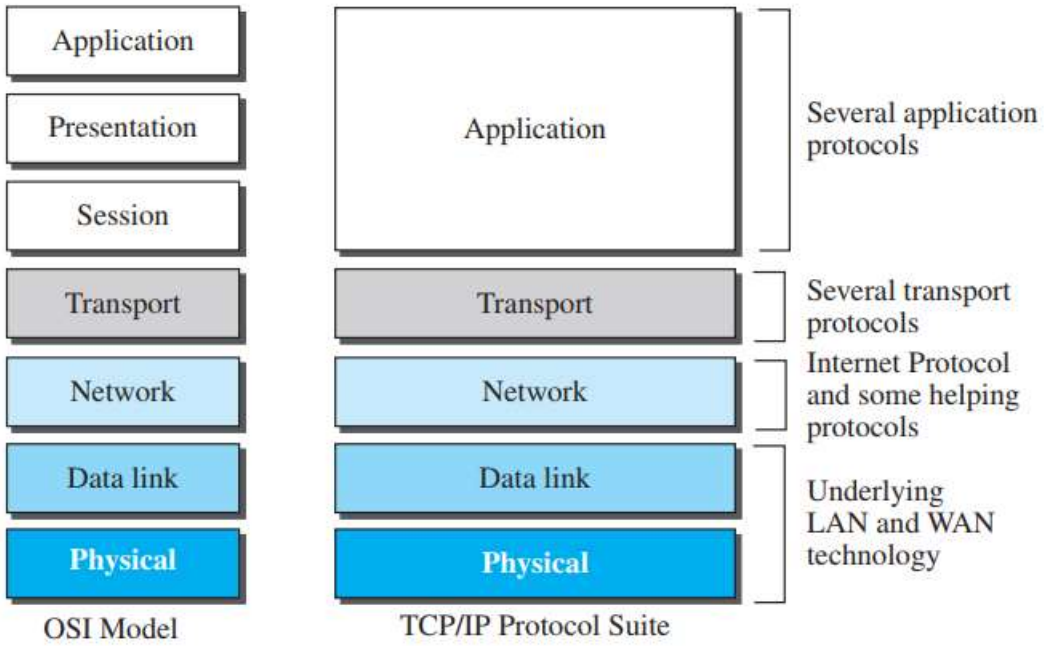
1) Application Layer
2) Presentation Layer
3) Session Layer
4) Transport Layer
5) Network Layer
6) Data Link Layer
7) Physical Layer

- Each layer has specific duties to perform and has to co-operate with the layers above & below it.



OSI vs. TCP/IP

1) When we compare the two models, we find that two layers, session and presentation, are missing from the TCP/IP protocol suite.
2) These two layers were not added to the TCP/IP protocol suite after the publication of the OSI model.
3) The application layer in the suite is usually considered to be the combination of three layers in the OSI model



Lack of OSI Model's Success

TCP/IP was already established

- By the time OSI was finished, the Internet was already running on TCP/IP.
- companies had invested a lot of time, money, and systems into TCP/IP.
- Replacing it would be too expensive and disruptive.

Some OSI layers were not fully defined

- OSI clearly listed what Session and Presentation layers should do…
- But it never fully defined how (no solid protocols, no real software).
- TCP/IP, on the other hand, had working protocols and real implementations.

Analog and Digital Data

Data can be analog or digital.
Analog data refers to information that is continuous;
Example,
1. an analog clock that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous.
2.the sounds made by a human voice, take on continuous values.

Digital data refers to information that has discrete states.
Example,
1. a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06

Chapter -3

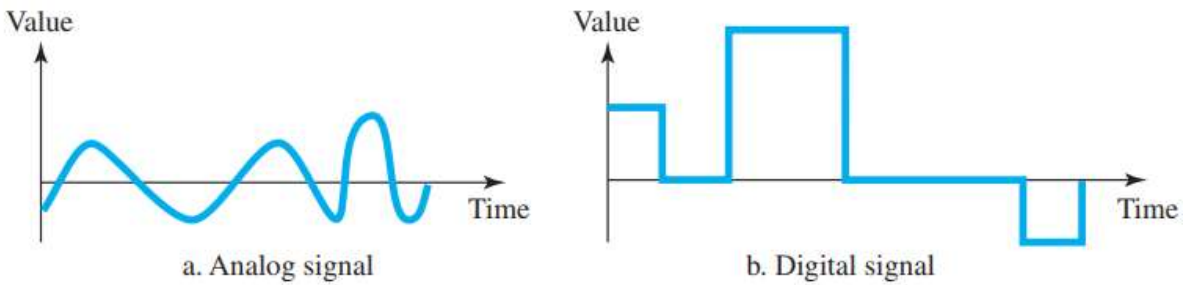
Analog and Digital Signal

signals can be either analog or digital.

- An **analog signal** has infinitely many levels of intensity over a period of time.

As the wave moves from value *A* to value *B*, it passes through and includes an infinite number of values along its path.

- A **digital signal,** on the other hand, can have only a limited number of defined values.



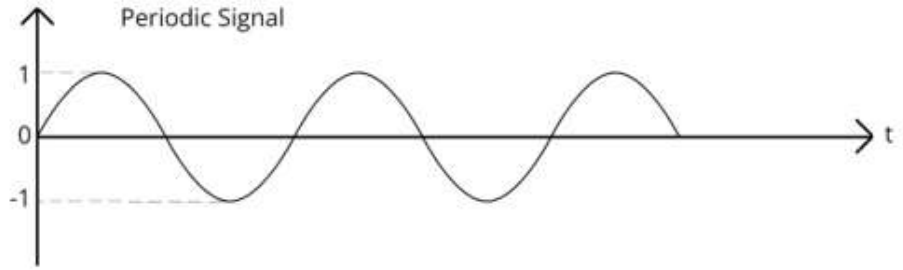
a. Analog signal b. Digital signal

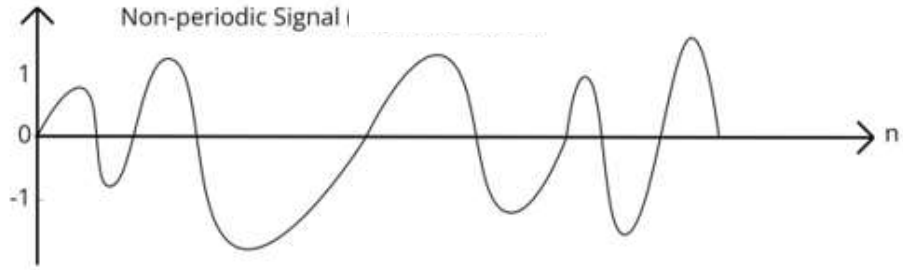
- The vertical axis shows the strength or value of a signal, while the horizontal axis shows time.
- An analog signal is a smooth, continuous curve that passes through an infinite number of points.
- A digital signal consists of sudden jumps between values, represented by vertical lines.

Periodic and Nonperiodic

Both analog and digital signals can take one of two forms: *periodic* or *nonperiodic*

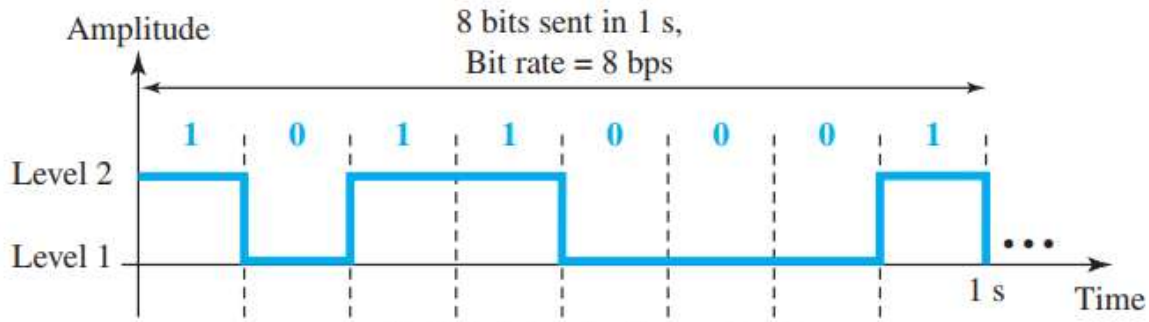
A **periodic signal** completes a pattern within a measurable time frame, called a **period,** and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a **cycle.**
A **nonperiodic signal** changes without exhibiting a pattern or cycle that repeats over time.
Both analog and digital signals can be periodic or nonperiodic.



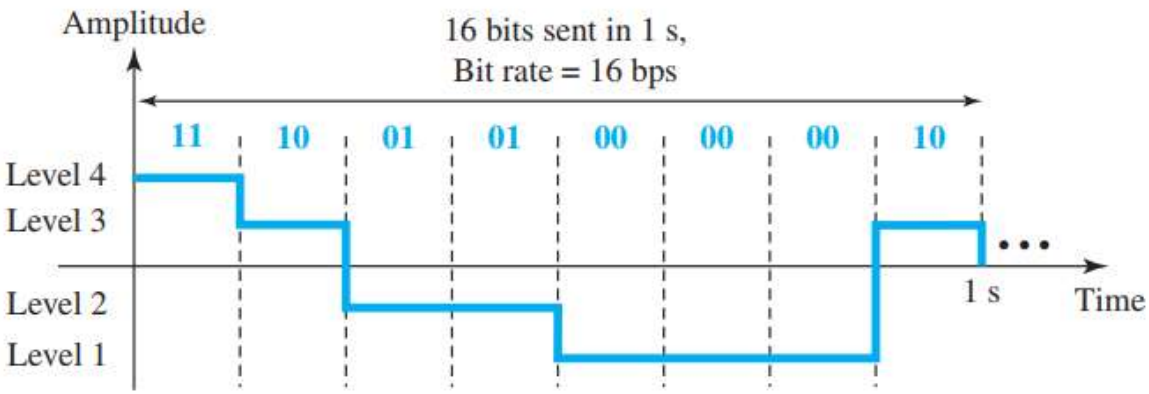


DIGITAL SIGNALS

- information can be represented by a digital signal.
- For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage.
- A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level. if a signal has *L* levels, each level needs log2 *L* bits. For this reason, we can send log24 = 2 bits in part b.



a. A digital signal with two levels



b. A digital signal with four levels

Bit Rate

The **bit rate** is the number of bits sent in 1s, expressed in **bits per second (bps)**

Problem

1. Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

100 * 24 * 80 * 8 = 1,536,000 bps = 1.536 Mbps

2. A digitized voice channel, is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

2 * 4000 * 8 = 64,000 bps = 64 kbps

3. What is the bit rate for high-definition TV (HDTV)?

Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9 .which means the screen is wider. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel. We can calculate the bit rate as

1920 * 1080 * 30 * 24 = 1,492,992,000 ≈ 1.5 Gbps

The TV stations reduce this rate to 20 to 40 Mbps through compression

Bit Length

The **bit length** is the distance one bit occupies on the transmission medium.

Bit length = propagation speed * bit duration
Bit length → The physical distance that **one bit** occupies on the transmission medium (meters).

- **Propagation speed** → How fast the signal travels through the medium (m/s).

- **Bit duration** → The time needed to send **one bit** (seconds).

Digital Signal as a Composite Analog Signal

A digital signal is a composite analog signal.

• A digital signal, in the time domain, comprises connected vertical and horizontal line segments.

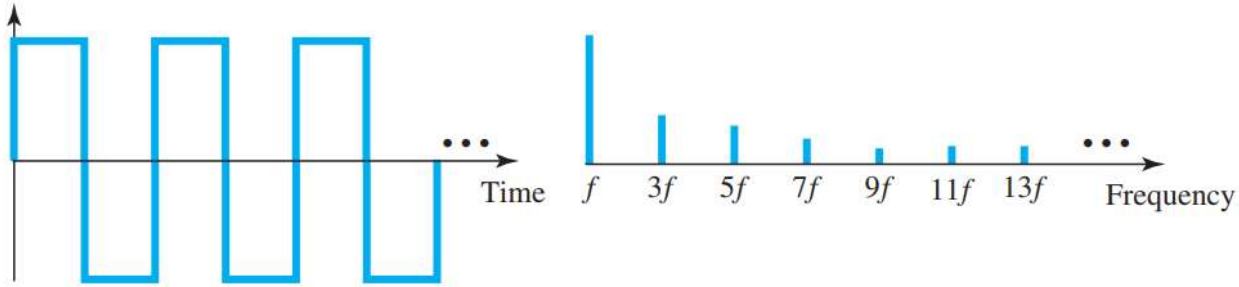
1) A vertical line in the time domain means a frequency of infinity (sudden change in time)

2) A horizontal line in the time domain means a frequency of zero (no change in time).

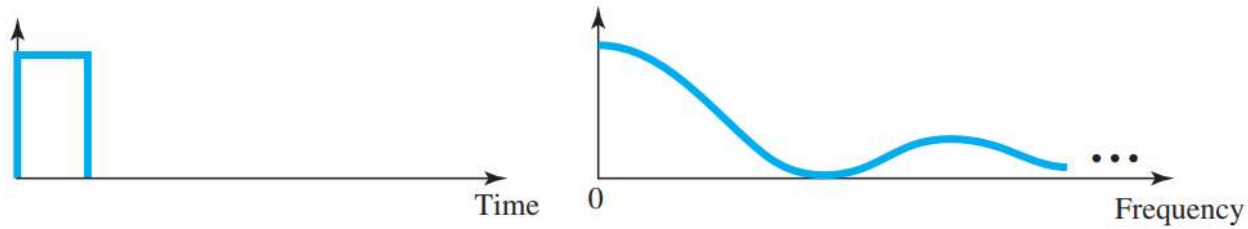
• Fourier analysis can be used to decompose a digital signal.

1) If the digital signal is periodic, the decomposed signal has a frequency domain representation with an infinite bandwidth and discrete frequencies (Figure a).

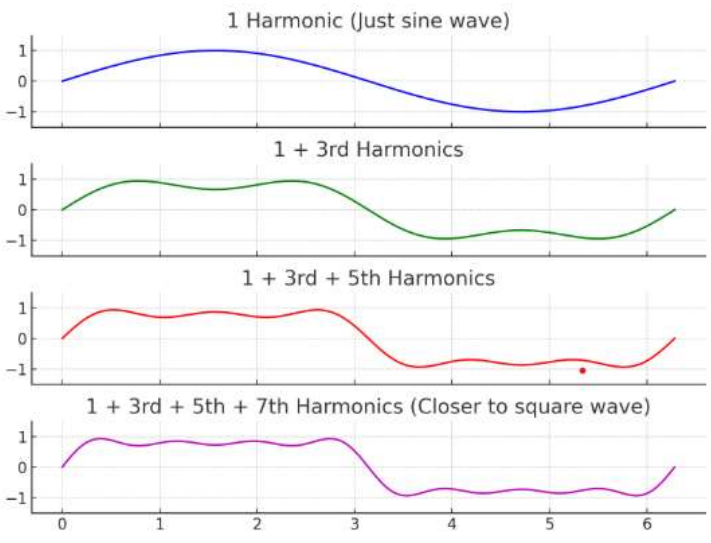
2) If the digital signal is non-periodic, the decomposed signal has a frequency domain representation with an infinite bandwidth and continuous frequencies (Figure b).



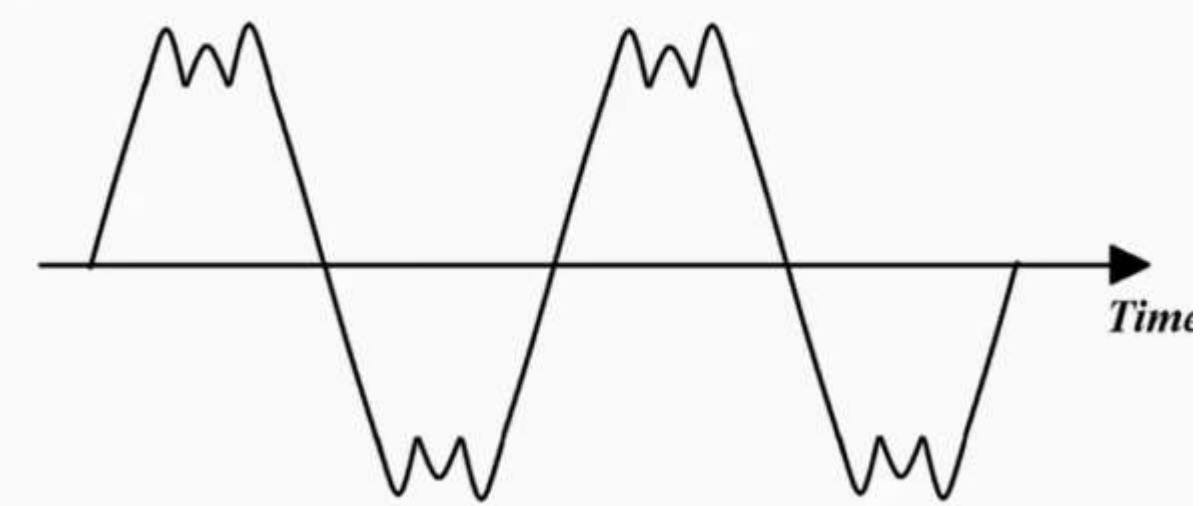
a. Time and frequency domains of periodic digital signal



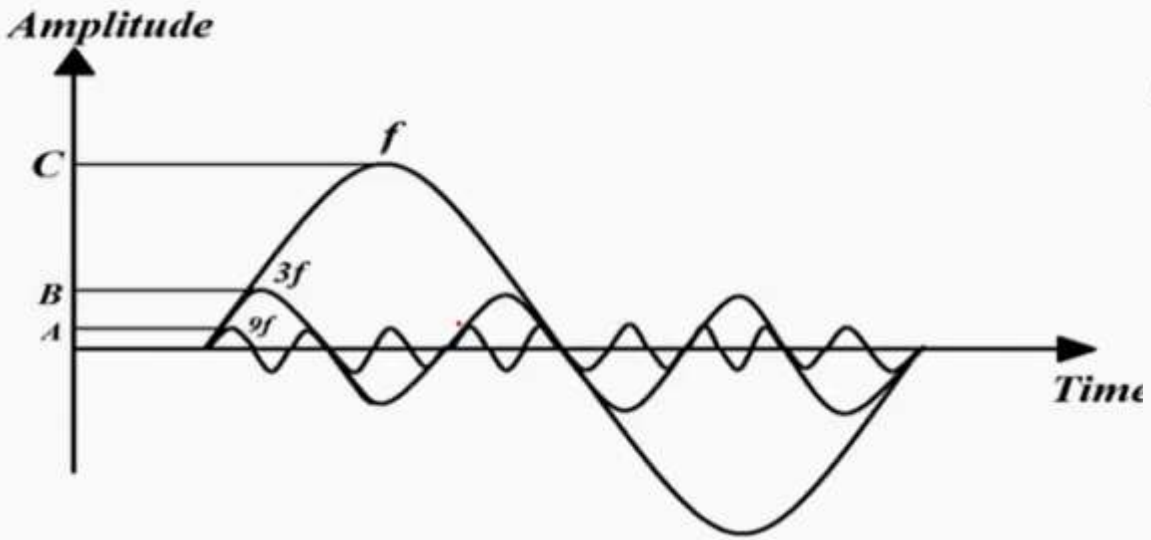
b. Time and frequency domains of nonperiodic digital signal

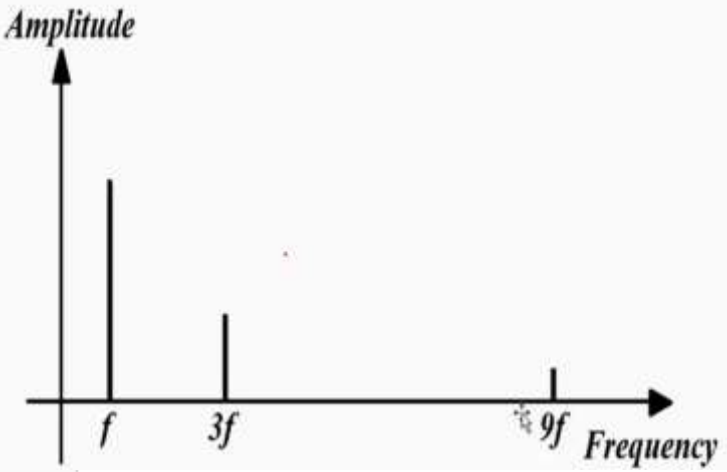


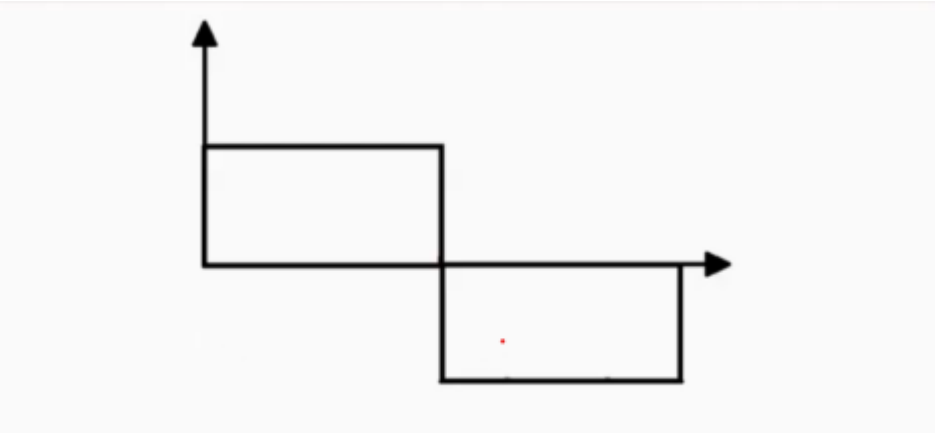
Composite Periodic Signal

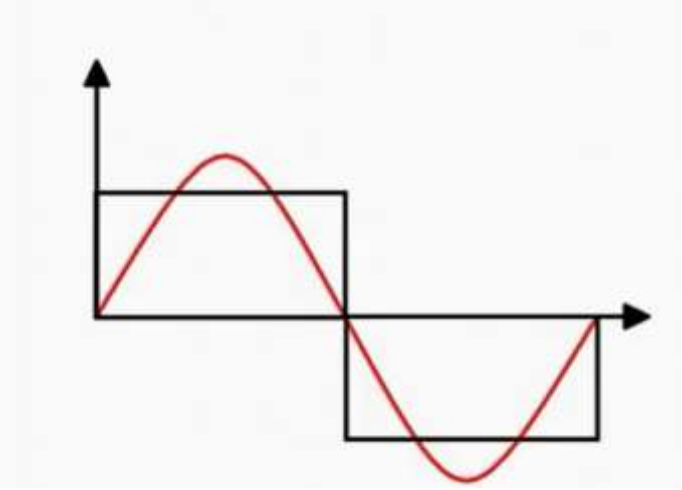


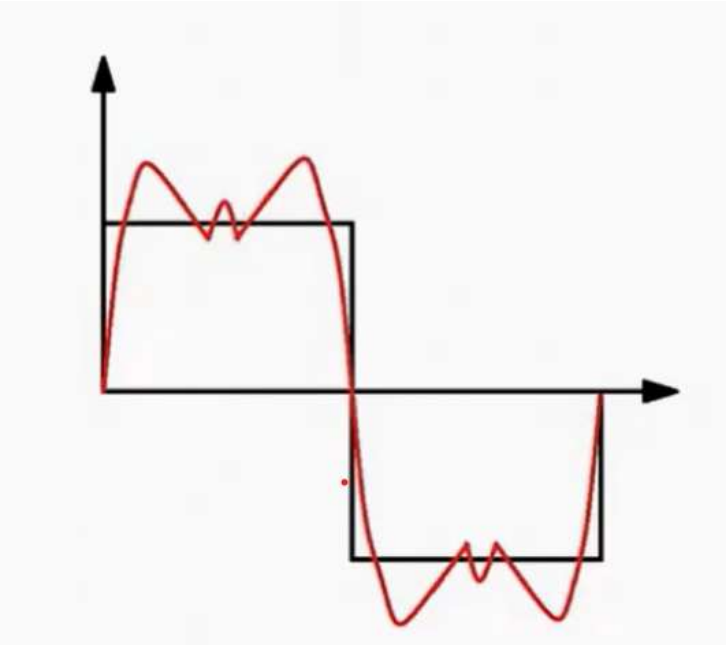
Decomposition of a Composite Periodic Signal

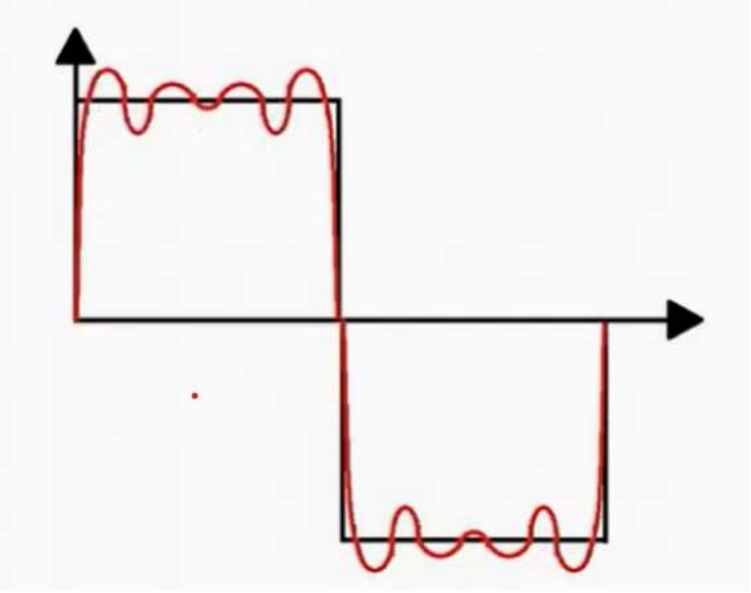


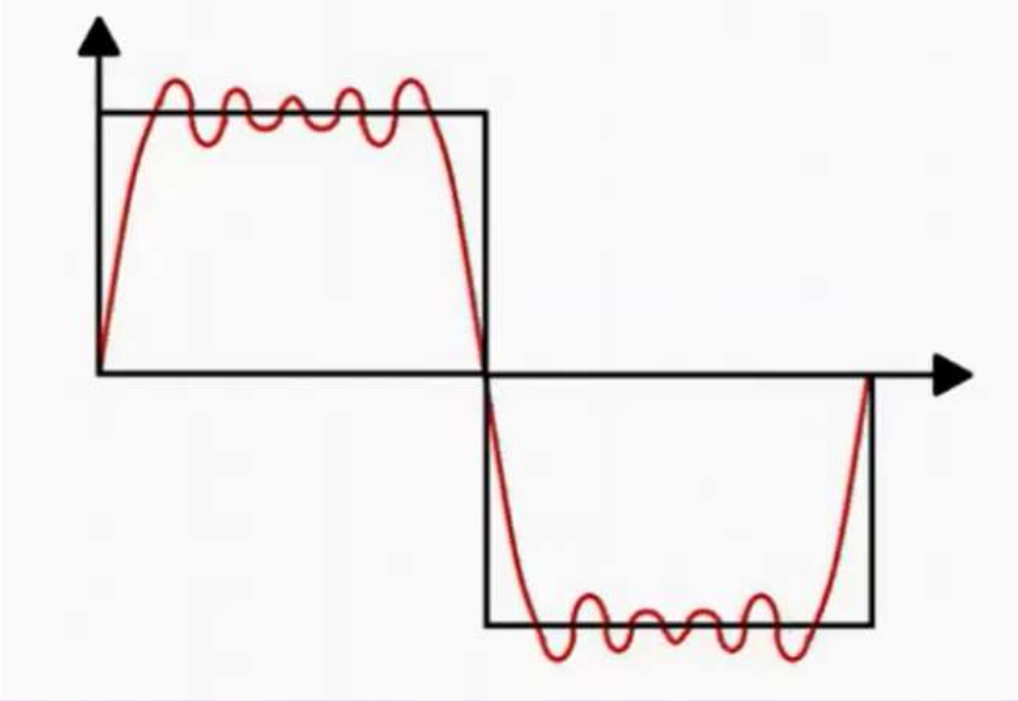




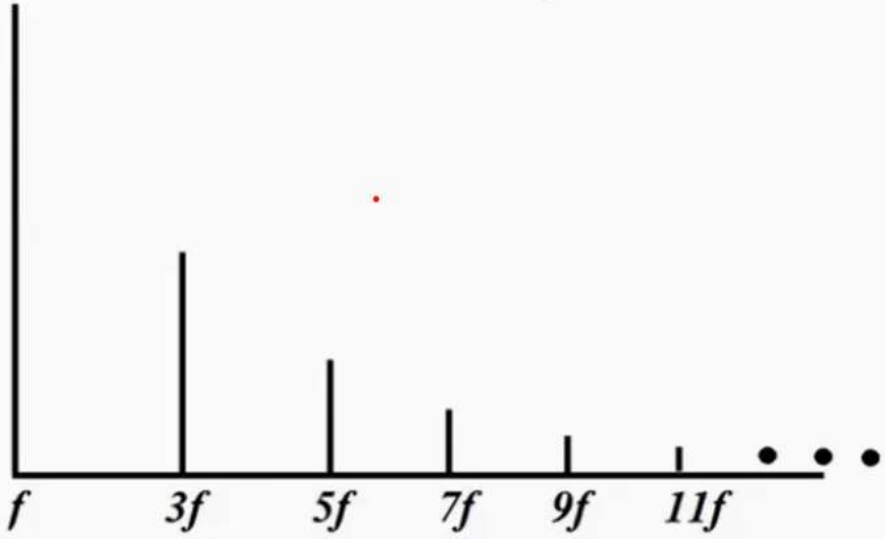








Frequency Domain Representation of a Digital Periodic Signal

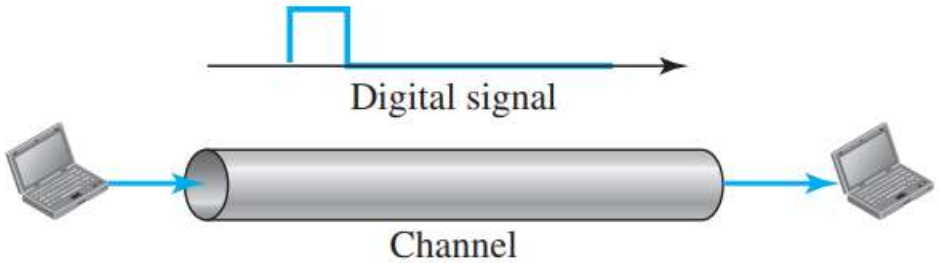


Transmission of Digital Signals

How can we send a digital signal from point *A* to point *B*?
transmit a digital signal by using one of two different approaches: baseband transmission or broadband transmission

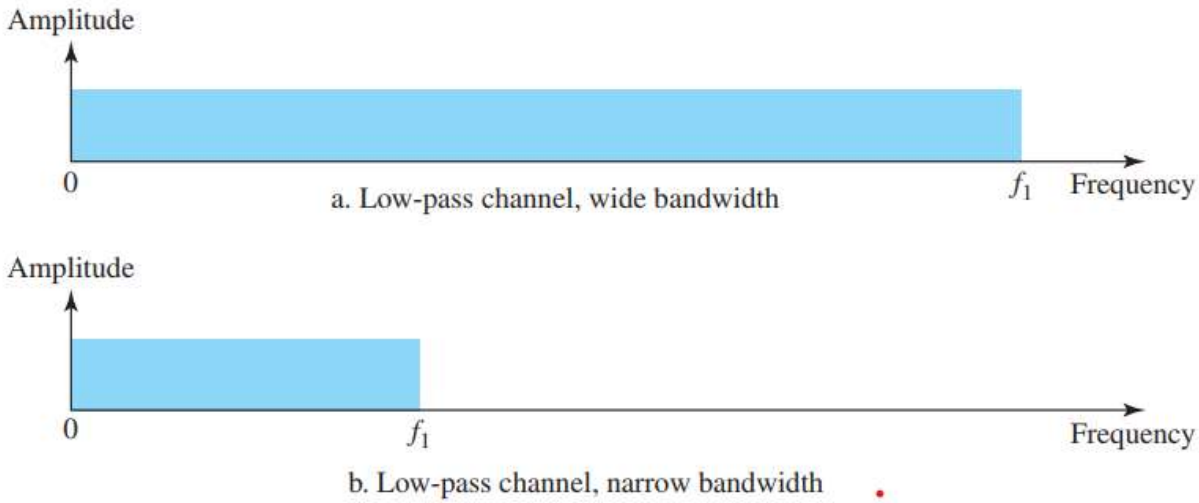
Baseband Transmission

Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.



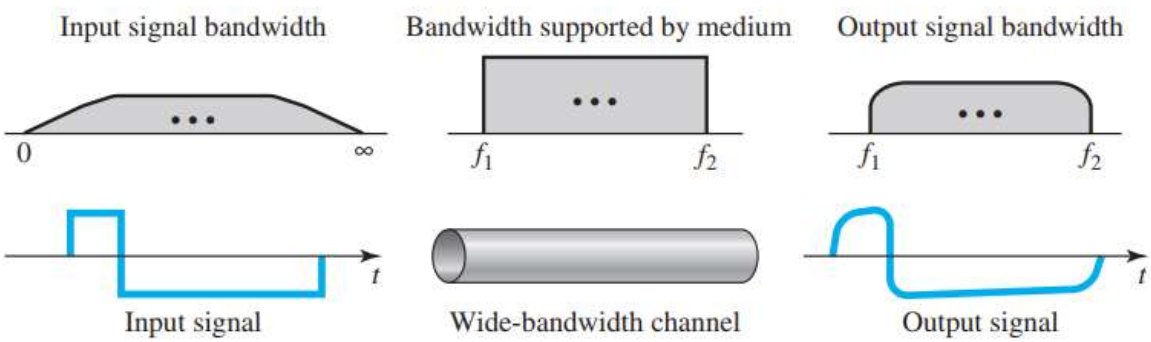
Baseband transmission requires that we have a **low-pass channel,** a channel with a bandwidth that starts from zero

Transmission of Digital Signals



Case 1: Low-Pass Channel with Wide Bandwidth

- To preserve the shape of a digital signal, we need to send the entire spectrum i.e. the continuous range of frequencies between zero and infinity.
- This is possible if we have a dedicated medium with an infinite bandwidth between the sender and receiver.
- If we have a medium with a very wide bandwidth, 2 stations can communicate by using digital signals with very good accuracy.
- Although the output signal is not an exact replica of the original signal, the data can still be deduced from the received signal.

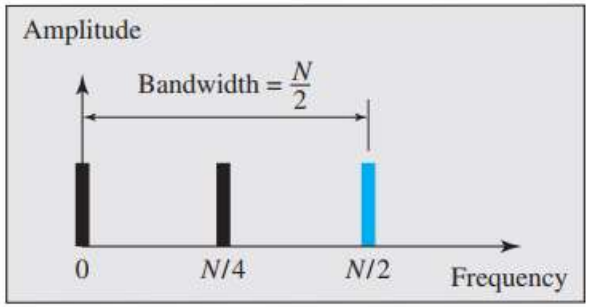


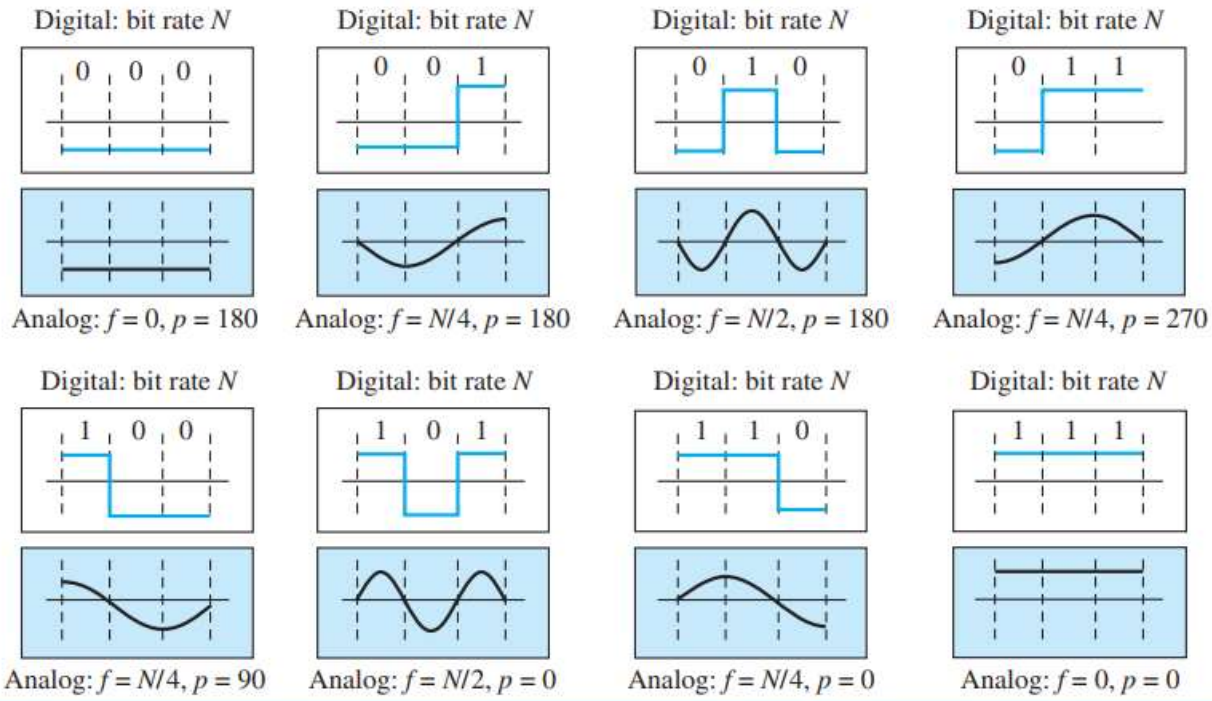
Case 2: Low-Pass Channel with Limited Bandwidth

In a low-pass channel with limited bandwidth, we approximate the digital signal with an analog signal. The level of approximation depends on the bandwidth available.

- ***Rough Approximation***
  Assume that we have a digital signal of bit rate N
- If we want to send analog signals to roughly simulate this signal, we need to consider the worst case, a maximum number of changes in the digital signal.
- This happens when the signal carries the sequence 01010101 . . or 10101010
- To simulate these two cases, we need an analog signal of frequency f = N/2.
- Let 1 be the positive peak value and 0 be the negative peak value.
- We send 2 bits in each cycle; the frequency of the analog signal is one-half of the bit rate, or N/2.
- This rough approximation is referred to as using the first harmonic (N/2) frequency. The required bandwidth is

$$\text{Bandwidth} = \frac{N}{2} - 0 = \frac{N}{2}$$


Amplitude
Bandwidth = $\frac{N}{2}$
0
$N/4$
$N/2$
Frequency

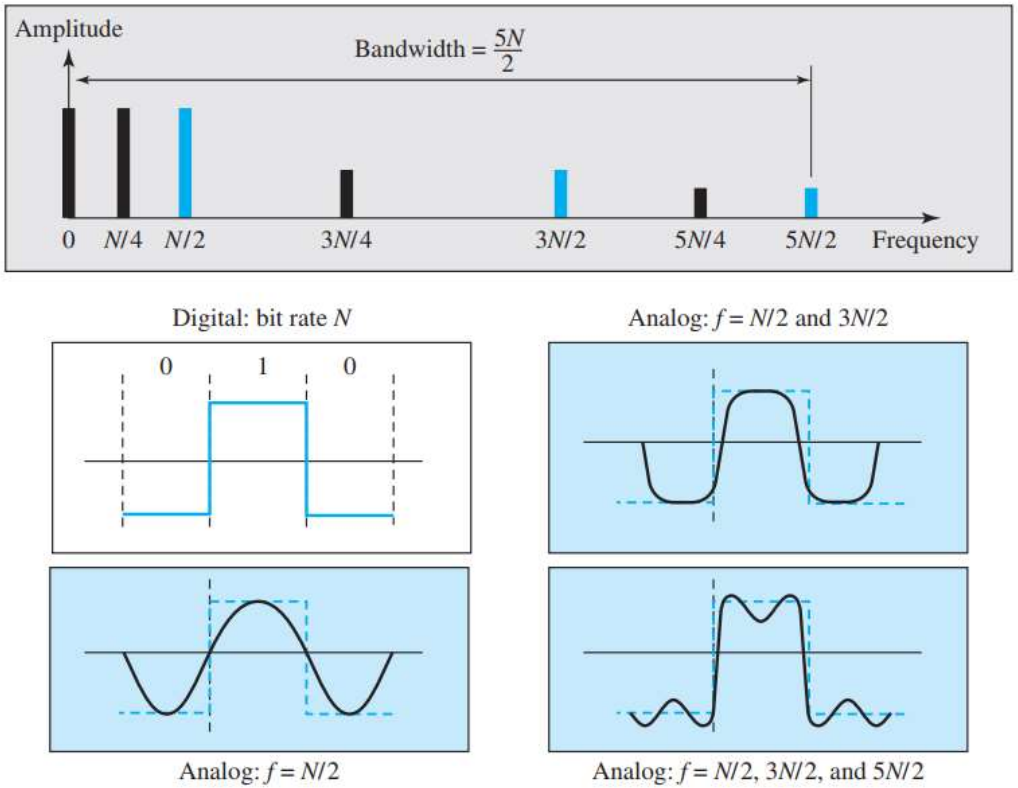

Digital: bit rate N
0 0 0
Analog: $f = 0$, $p = 180$
Digital: bit rate N
0 0 1
Analog: $f = N/4$, $p = 180$
Digital: bit rate N
0 1 0
Analog: $f = N/2$, $p = 180$
Digital: bit rate N
0 1 1
Analog: $f = N/4$, $p = 270$
Digital: bit rate N
1 0 0
Analog: $f = N/4$, $p = 90$
Digital: bit rate N
1 0 1
Analog: $f = N/2$, $p = 0$
Digital: bit rate N
1 1 0
Analog: $f = N/4$, $p = 0$
Digital: bit rate N
1 1 1
Analog: $f = 0$, $p = 0$

Better Approximation

To make the shape of the analog signal look more like that of a digital signal, we need to add more harmonics of the frequencies

We can increase the bandwidth to 3N/2, 5N/2, 7N/2, and so on.

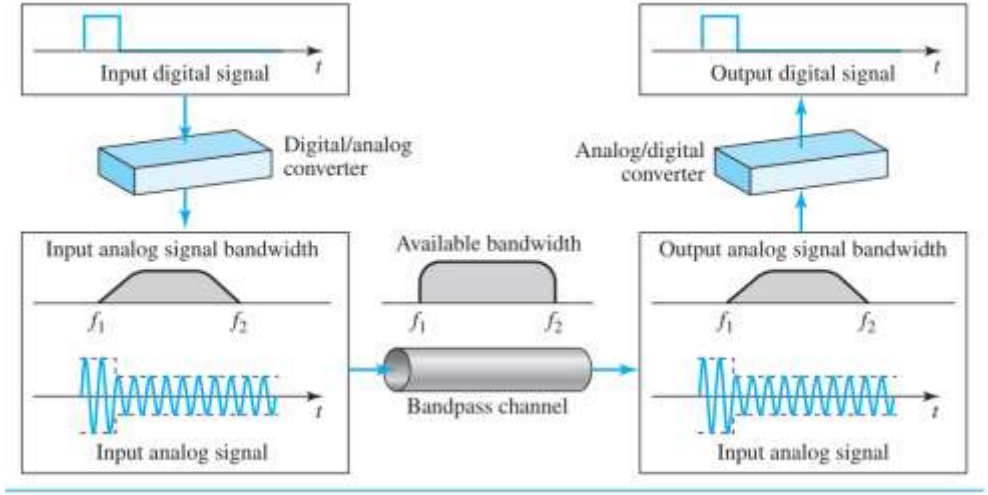
In baseband transmission, the required bandwidth is proportional to the bit rate; If we need to send bits faster, we need more bandwidth.



Broadband Transmission (Using Modulation)

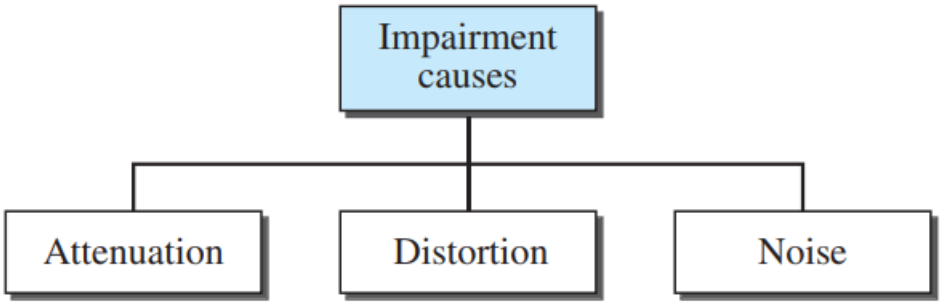
Broadband transmission or modulation means changing the digital signal to an analog signal for transmission.

- Modulation allows us to use a bandpass channel.
- Bandpass channel means a channel with a bandwidth that does not start from zero.
- This type of channel is more available than a low-pass channel.



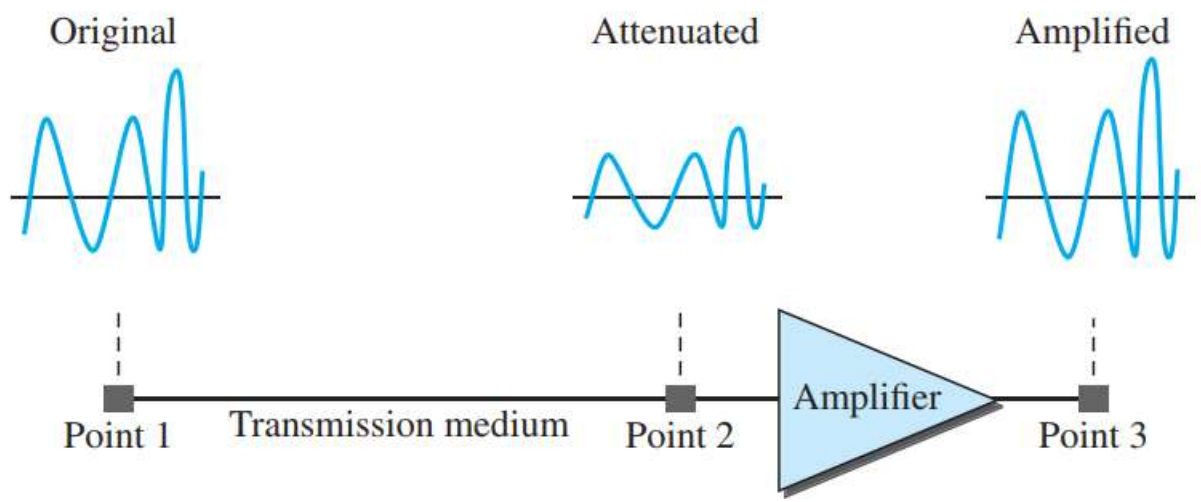
TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise



Attenuation

Attenuation means a loss of energy.
When a signal, travels through a medium, it loses some of its energy ,that is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat. To compensate for this loss, amplifiers are used to amplify the signal.



Decibel

Decibel

To show that a signal has lost or gained strength, engineers use the unit of the decibel.

The **decibel (dB)** measures the relative strengths of two signals or one signal at two different points. the decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$\mathbf{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

Variables P_1 and P_2 are the powers of a signal at points 1 and 2, respectively.

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that $P_2 = \frac{1}{2} P_1$. In this case, the attenuation (loss of power) can be calculated as

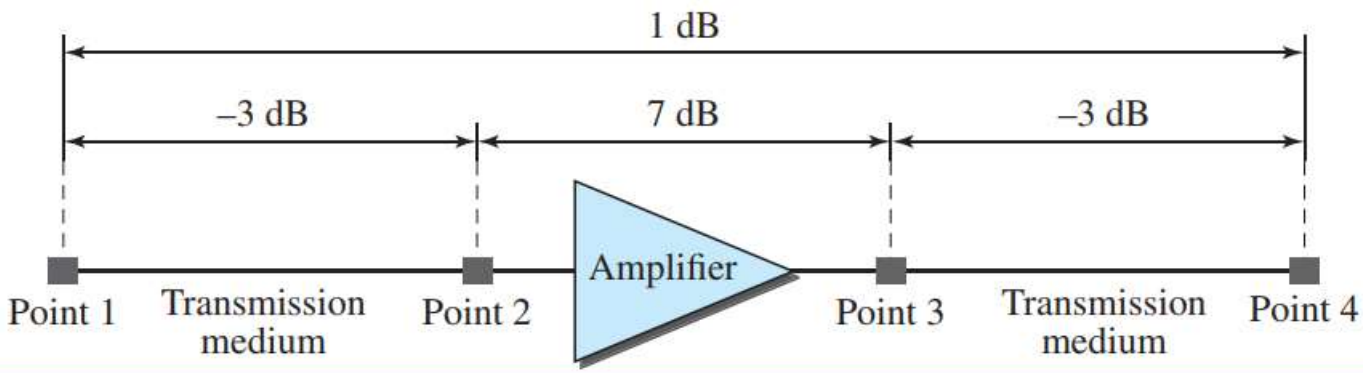
$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

a signal travels from point 1 to point 4. The signal is attenuated by the time it reaches point 2. Between points 2 and 3, the signal is amplified. Again, between points 3 and 4, the signal is attenuated. We can find the resultant decibel value for the signal just by adding the decibel measurements between each set of points.



In this case, the decibel value can be calculated as

$$\mathbf{dB = -3 + 7 - 3 = +1}$$

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dB_m and is calculated as $dB_m = 10 \log_{10} P_m$, where P_m is the power in milliwatts. Calculate the power of a signal if its $dB_m = -30$.

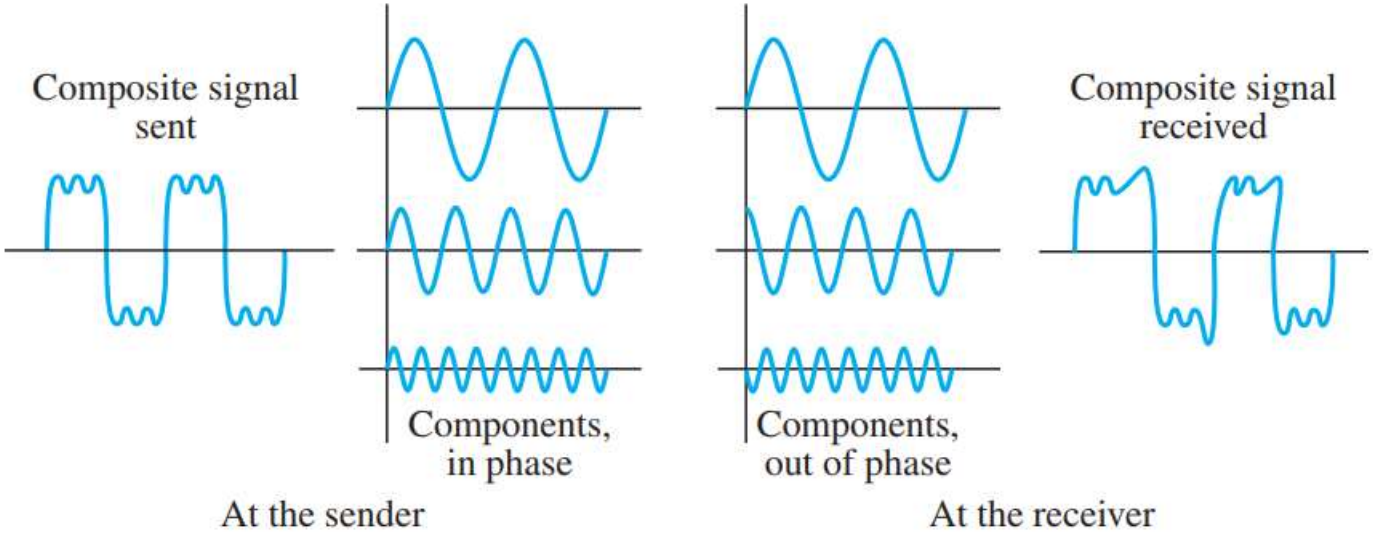
Solution

We can calculate the power in the signal as

$$\mathbf{dB_m = 10 \log_{10} \longrightarrow dB_m = -30 \longrightarrow \log_{10} P_m = -3 \longrightarrow P_m = 10^{-3}\,mW}$$

Distortion

- **Distortion** means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- Each signal component has its own propagation speed (see the next section) through a medium and, therefore, its own delay in arriving at the final destination.
- signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same.



Noise

Noise in Communication

Noise (unwanted Signals) distorts or corrupts the original signal, making it harder for the receiver to understand.

Types of Noise

Thermal Noise

Caused by the random motion of electrons inside the wire.

Induced Noise

Comes from external devices like motors, household appliances, or other machinery.

These devices generate electromagnetic waves, which get picked up by nearby cables

Crosstalk

Happens when one communication channel **leaks** into another.

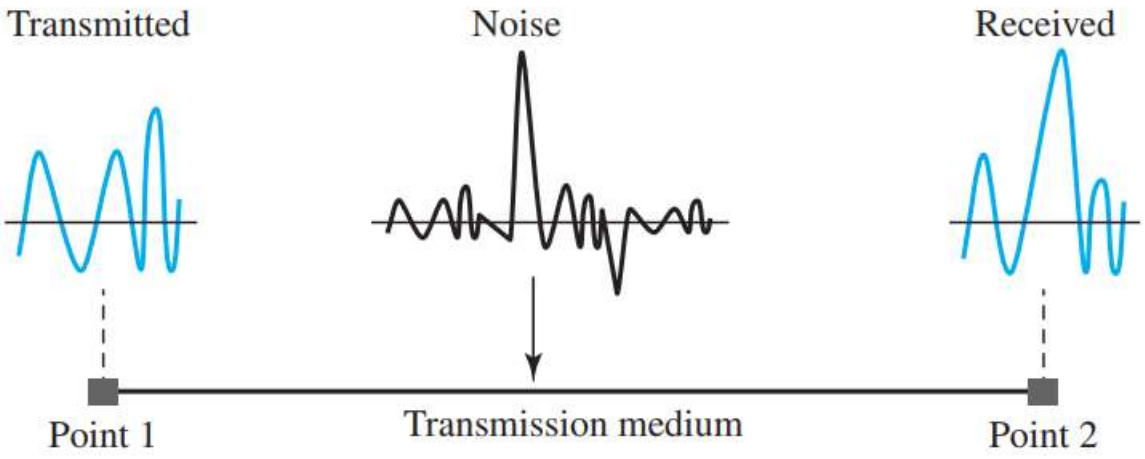
Example: you hear another person's phone conversation faintly in the background.

This occurs because wires next to each other act like antennas — one transmits unintentionally, the other receives.

Impulse Noise

Sudden, very short bursts of high energy.

Caused by things like lightning, power lines, or switching equipment.

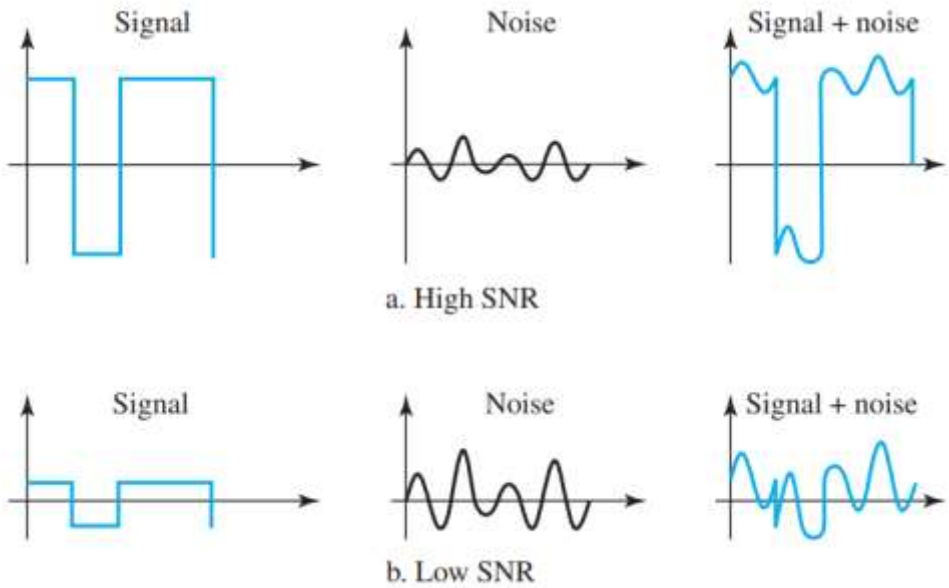


Signal to Noise ratio

The signal-to-noise ratio is defined as

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

average signal power and the average noise power has to be considered because these may change with time



SNR is actually the ratio of what is wanted (signal) to what is not wanted (noise). A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise.

The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB}?

Solution

The values of SNR and SNR_{dB} can be calculated as follows:

$$\mathbf{SNR = (10{,}000\ \mu w) / (1\ \mu w) = 10{,}000 \qquad SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40}$$